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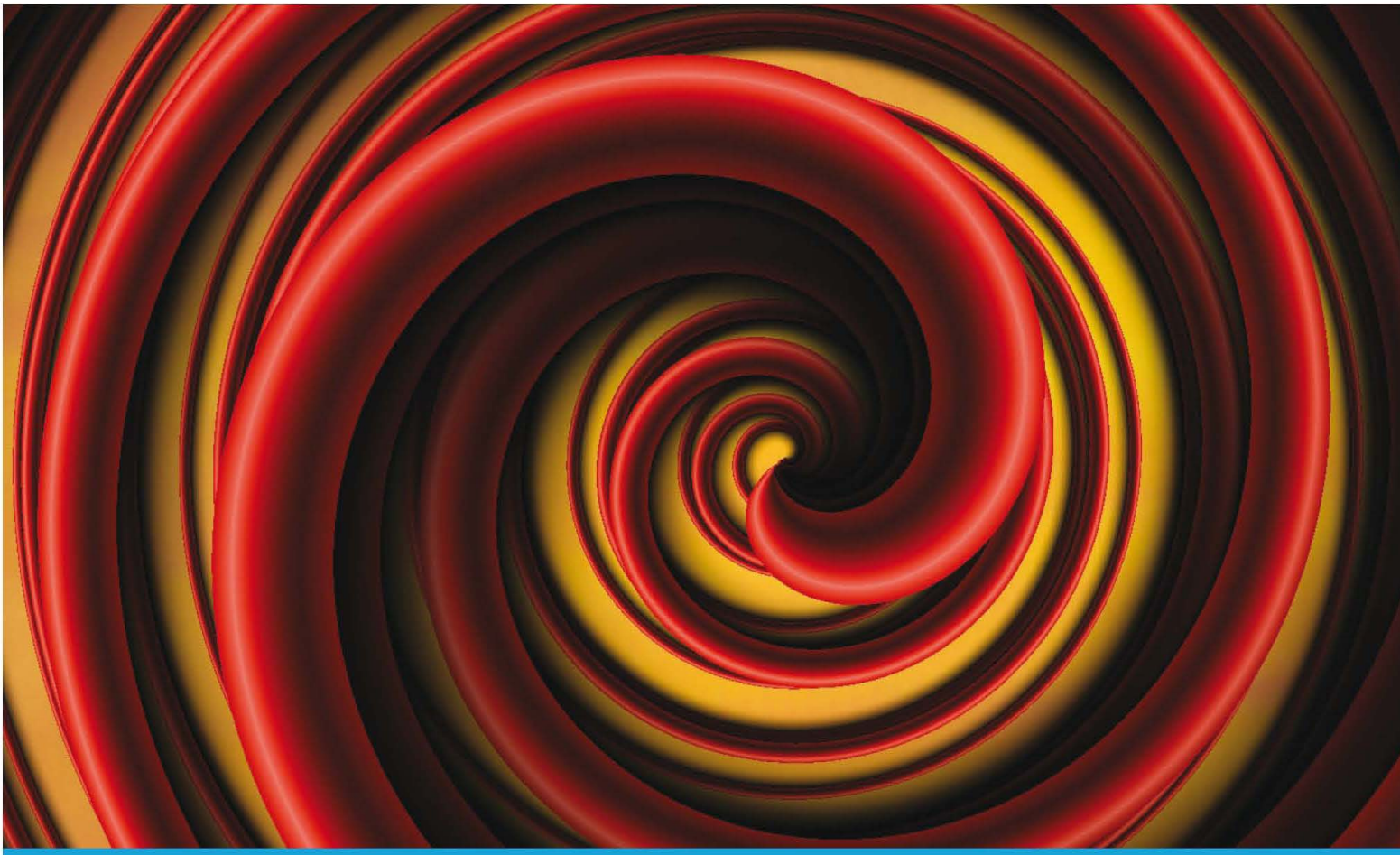

Microsoft Dynamics AX 2012 Security How-To

Secure your Dynamics AX 2012 environment with simple and practical steps

Ahmed Mohamed Rafik Moustafa

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When he isn't focusing on his career, Ahmed enjoys his favorite sports such as football, swimming, and tennis. He is also a good piano player and as you can see, he tries to keep a balance in his life through his diverse interests because he enjoys living life with joy and passion.

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I dedicate this book to the memory of my father, and a special dedication to my mother, who has supported me my whole life as she is the one for whom I live.

To my uncle, Dr. Ahmed Hegazy, who inspired me to succeed and work for the sake of God and His Prophet Mohamed (peace be upon him).

Also, I cannot possibly overlook the motivation provided by my sisters, who continue to enthusiastically encourage me to make a significant impact in this world.

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Table of Contents

Preface 1

Microsoft Dynamics AX 2012 Security How-To 5

Security architecture in Dynamics products 5
Application security fundamentals 7
AX 2012 security framework 9
Creating a new user (Must know) 12
Importing users from Active Directory 15
Creating a claim user (Should know) 21
Setting up user (Should know) 22
Setting up user options (Should know) 26
Setting up user relations (Should know) 28
Adding a user to a predefined profile (Should know) 29
Granting a user access to a specific company (Must know) 32
Tracking user-related information (Must know) 34
Assigning users to different roles (Must know) 37
Adding users to the different roles (Should know) 39
Adding a security role (Become an expert) 43
Adding privileges to an existing role (Become an expert) 47
Securing the Organization Model (Become an expert) 50
Developing an extensible data security policy (Become an expert) 53
Deploying changes in Enterprise Portal using AOT (Become an expert) 56
Creating permissions in AOT (Become an expert) 57

Preface

Welcome to the Microsoft Dynamics AX 2012 Security book, where we take you on a journey starting from the big changes happened in Microsoft Dynamics AX 2012, ending with the practical steps to make the necessary security setup illustrated with snapshots and figures that will guide you through your ERP implementation process.

What this book covers

Creating a new user (Must know): This recipe explains the steps necessary to create a new user on Microsoft Dynamics AX 2012 and modify user-related information.

Importing users from Active Directory: This recipe explains the steps necessary to import the created users, from Active Directory to Microsoft Dynamics AX 2012, in a simple and easy way.

Creating a claim user (Should know): This recipe explains the steps required to create users who are not in an organization's Active Directory, such as vendors and customers.

Setting up user (Should know): This recipe explains the setup required to assign users to user groups on Microsoft Dynamics AX 2012.

Setting up user options (Should know): This recipe explains the steps required to set up user options on Microsoft Dynamics AX 2012.

Setting up user relations (Should know): This recipe explains the steps required to determine user relations with your company—employee, customer, vendor, or prospect.

Adding a user to a predefined profile (Should know): This recipe explains the steps necessary to assign a user to a predefined user profile.

Granting a user access to a specific company (Must know): This recipe explains the steps required to assign or unassign the users access to specific organizations on Microsoft Dynamics AX 2012.

Tracking user-related information (Must know): This recipe explains the steps necessary to track the user's login and logoff records by using the user's log and online time inquiry.

Assigning users to different roles (Must know): This recipe explains the steps necessary to know how users can be assigned to different security roles on Microsoft Dynamics AX 2012.

Adding a security role (Become an expert): This recipe explains the interaction of the user with AOT to know how to add a security role and make the necessary setup.

Adding privileges to an existing role (Become an expert): This recipe explains the steps necessary to add privileges or process cycles to an existing role in a simple and easy way.

Securing the Organization Model (Become an expert): This recipe explains the steps necessary to know how to limit the user's access according to the Organization Model hierarchy.

Developing an extensible data security policy (Become an expert): This recipe explains the steps required to create a data security policy based on the primary relation with a specific table.

Deploying changes in Enterprise Portal using AOT (Become an expert): This recipe helps us understand how to apply the changes that occurred on AOT to Enterprise Portal.

Creating permissions in AOT (Become an expert): This recipe helps us learn how to create a permission to a specific security role using the AOT form.

What you need for this book

You will need to understand each and every section well and then practice the examples covered in this book by following the step-by-step approach. You also need to run Microsoft Dynamics AX 2012 Virtual Machine and perform every step so that you understand the execution properly.

Who this book is for

This book is for consultants, implementers, project managers, IT managers, and developers.

Conventions

In this book, you will find a number of styles of text that distinguish between different kinds of information. Here are some examples of these styles, and an explanation of their meaning.

New terms and **important words** are shown in bold. Words that you see on the screen, in menus or dialog boxes for example, appear in the text such as "clicking on the **Next** button moves you to the next screen".

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Microsoft Dynamics AX 2012 Security How-To

Welcome to Microsoft Dynamics AX Security 2012, where I will take you on a journey that starts with the big changes happening in Microsoft Dynamics AX 2012 and ends with the practical steps that are needed to make the necessary security setup, illustrated with snapshots and figures, that will guide you through the ERP implementation process.

Security architecture in Dynamics products

Before going into the security architecture, we have to be aware of security concepts or know the security terms that we should consider while implementing the security on Microsoft Dynamics AX. We will focus on the three main concepts of security:

- **Attack surface**: If you are always worried about your data or your code being stolen, destroyed, or damaged by an unauthorized user, you have to perform the necessary procedures and follow certain policies to secure your data against attack surface.

  By disabling ports and services you can reduce the risk of an attack from the Internet and network, or limit access to computer resources by enabling hardware and software security features.

- **Defence in depth**: Defence in depth means building a lot of security defences that limit unauthorized users accessing your data. For example, we can build a lot of firewalls to make it harder to attack for a malicious user. It is also recommended to physically secure your server in such a way that one cannot enter the server room without a special ID. You can also use CCTV cameras to record what is happening in the server room.

Obtain Internet access for users through the firewall to avoid bad requests to the client computer. Also install antivirus software on your local machines to reduce the chances of being attacked by viruses.

- **Least privileges**: Can you just give the users the privileges that enable them to do only their job? You just want the users to do their entire processes and tasks with few privileges, without giving them access to the other operations and information they do not require.

You must be aware of these concepts in order to secure your data from being hacked, corrupted, or damaged.

Securing the business processes

In addition to system security, professionals and system consultants need to pay attention to enforcing best practices for securing business operations such as implementing proper role-based clients and limited access to trade agreements.

Let's go through the security architecture in the Microsoft Dynamics AX product. The security architecture in the Microsoft Dynamics AX product consists of the infrastructure security and application security.

- **Infrastructure security** – the Microsoft Dynamics AX infrastructure is built on the following features:
  - Active Directory services
  - Integrated windows authentication
  - Computer networking
  - Secured servers machine
- **Application security** – application security has the same features as those listed for Infrastructure Security and includes the following additional features:
  - Active Directory users added to Microsoft Dynamics AX and granted permissions to use the application
  - Domains that are groups of the company accounts in Dynamics AX
  - Record-level security to restrict or permit users to access specific fields and tables
  - Security keys that allow users to access specific forms, reports, or menus

We saw the security architecture in Microsoft Dynamics Products and now we are going to see the Application Security fundamentals in Microsoft Dynamics AX.

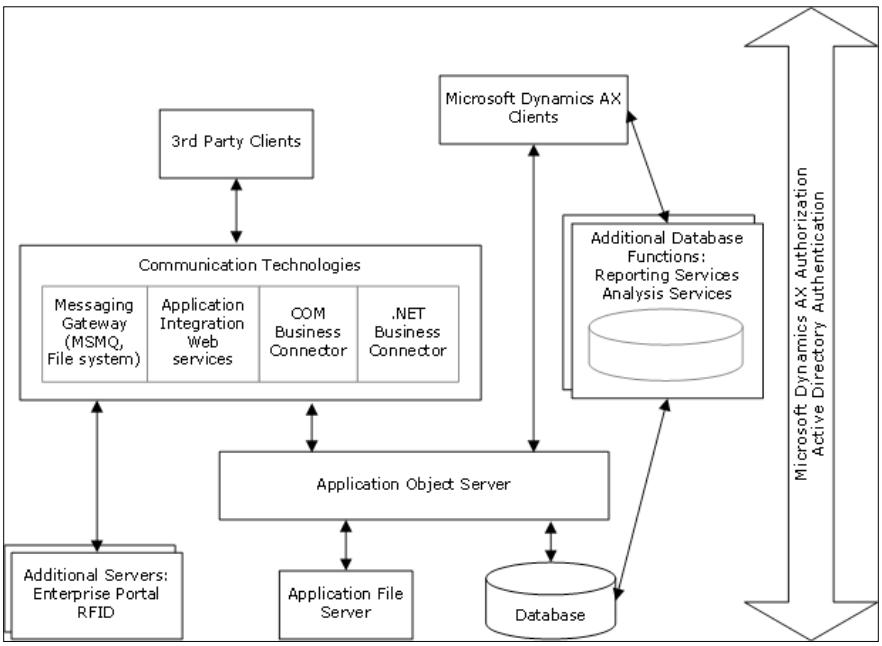
Application security fundamentals

Before securing the application in Microsoft Dynamics AX, we have to secure the environment.

The Microsoft Dynamics AX system requires Active Directory services, a network with firewall configured with one or two domain controllers, for the purpose of installing and configuring Enterprise Portal.

Microsoft Dynamics AX is based on the client/server model, so it is necessary to know how to secure both the client that makes the service request and the server that fulfills the request.

The following figure shows the high-level architect of Dynamics AX:



Source: Microsoft MSDN

Microsoft Dynamics AX server security

We are going to take a look at the necessary instructions that should be considered when securing the Dynamics AX server:

- **Application file server**: The application files should be restricted to the application object server domain account and the administrator.
- **Database server**: The database server should be secured using the recommended SQL server security practices.

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- **Application object server (AOS)**: AOS should be restricted to the log directory (`installationdirectory\log`) for only the AOS account directory and the administrators. For the domain account of the AOS, the account should be granted rights as follows:
  - Log on as a service
  - No positive privileges should be granted.
- **Enterprise portal**: Securing an enterprise portal begins with securing **Microsoft Internet Information Services** (**IIS**) by using **Secure Sockets Layer** (**SSL**) so that it is the only mechanism for a secure Internet login.
- **Business connector**: This function is used by Enterprise Portal for external users, and the recommended account setup is as follows:
  - Password doesn't expire
  - No interactive logon rights

  It is better to have a complex password for the .NET business connector.

Microsoft Dynamics AX client security

We are going to take a look at the necessary instructions that should be considered when securing the Dynamics AX client:

- **Client desktop**: Set up the appropriate permissions for the shared folders and drivers
- **Business connector proxy user**: This function is used by Enterprise Portal for external users, and the recommended account setup is as follows:
  - Password doesn't expire
  - No interactive logon rights
- Other procedures must be specified for the computer accounts that are used to communicate with the server connection:
  - Between the database server and the local system, the MSSQL service must be running on the client machine (local system)
  - The domain account or network service account that is associated with the AOS service must communicate with the database server and grant a local user the appropriate rights as a user in the database

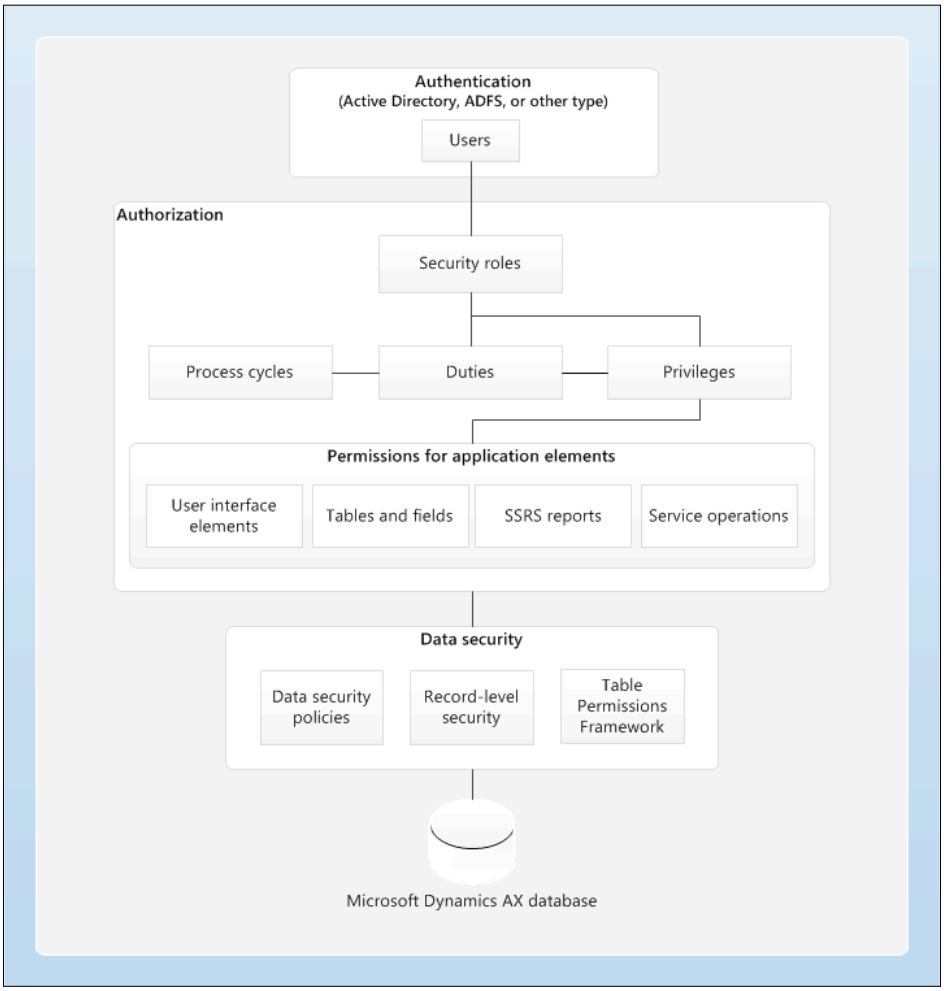
We have seen the high-level architecture of Dynamics Products and also the application security fundamentals. So, are you ready for the new security features in Microsoft Dynamics AX 2012? Let's go!

AX 2012 security framework

We are going to take a look at the following changes that have taken place in the security framework of Microsoft Dynamics AX 2012:

- Role-based security
- Extensible data security framework
- Server enforcement of security
- Custom authentication

The following figure shows the security architecture of Dynamics AX:



Source: Microsoft Tech-Net

Role-based security

In earlier versions of Microsoft Dynamics AX, the IT administrator wasted a lot of time and effort in managing the application security by creating user groups. By using the role-based security that Microsoft Dynamics AX 2012 offers, managing the security within the application will be easier and aligned with your business as it provides predefined roles found in business.

It also provides predefined business-related duties that are assigned with roles and matched with the users assigned to a specific role.

Extensible data security framework

Any data located in any table can be secured easily. The new framework helps you easily create policies based on data to secure your data in different tables.

You can specify which data you want the user to access—present, past, and even future records. Besides that, you can determine the level of access based on an effective date, so you may want to give the user access to the present data and not to the past data, based on a determined date. For your information, record-level security is still being used in Microsoft Dynamics AX 2012.

Server enforcement of security

In earlier versions of Microsoft Dynamics AX, authorizations were performed on the client's machine, but in Microsoft Dynamics AX 2012, authorizations are performed on the server. There is an addition made in **Table Permission Framework** (**TPF**) that will not only deny users access to full records, but also deny access for any field that was not supported before. This increases the consistency of security between client types and shifts the security load to the server.

Custom authentication

It is already known that user authentications are based on Active Directory, so all users must be domain users. But if we want to talk about the external users, in earlier versions of Microsoft Dynamics AX you had to create the external users as domain users, so in the new version of Microsoft Dynamics AX 2012, external users will no longer require domain accounts to be able to access Microsoft Dynamics AX 2012. There are a lot of methods besides Active Directory; these methods simplify the administration and maintenance required for external users and is called **pluggable authentication**.

Pluggable authentication

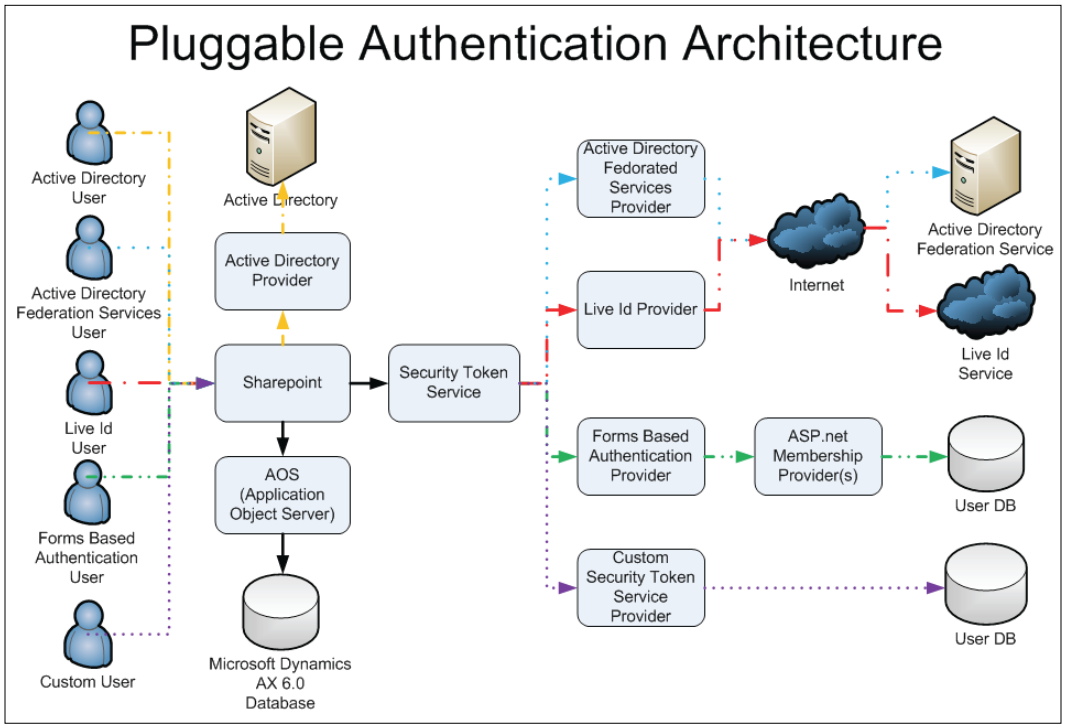
Pluggable authentication is new in Microsoft Dynamics AX 2012, and is used to allow access to Enterprise Portal to users who are not part of Active Directory. There are other types of authentication in addition to Active Directory, and they are as follows:

- **Active Directory Federated Service (ADFS)**: This allows users who are associated with an external Active Directory to access Enterprise Portal.

  When an ADFS user is removed from the external Active Directory, that user does not have access to Enterprise Portal.

- **Forms-based authentication**: This allows users to authenticate against a custom database of users.
- **Live ID**: Users can authenticate themselves to Enterprise Portal by using their Windows Live ID.

The following figure shows the pluggable authentication architecture for Dynamics AX 2012:



Source: What's New - Technical in Microsoft Dynamics® AX 2012 for Implementation

In Dynamics AX, the administration and authorization are based on the users, user groups, domain, and access rights. The new concept of security in AX 2012 has made the security easier to manage and provides a lot of benefits that enable the creation and editing of the user's duties and responsibilities more efficiently.

Users are assigned to roles based on their responsibilities in the organization and their participation in the business processes. So, when you make the necessary setup for editing or creating roles, the role assignment and privileges are updated based on changes in business data. Roles apply across all companies and not for a single organization, so the administrator no longer has to maintain separate user groups for each company.

Administrators and auditors can obtain information more easily about every user and their duties, responsibilities, and permissions; also, they can adjust the security for each user, based on their job in the company, with easier and faster steps.

Microsoft Dynamics AX 2012 provides better security audit trails and also a lot of security reports that help the auditors and administrators receive information about each user's privileges, duties, and permissions more easily. The new built-in feature **Segregation of Duties** makes sure no user has access to conflicting tasks or permissions.

In this section we will cover the following topics:

- Creating a new user
- Importing users from Active directory
- Creating a claim user
- Setting up user
- Setting up user options
- Setting up user relations
- Adding a user to a predefined profile
- Granting a user access to a specific company
- Tracking user-related information
- Assigning users to different roles
- Adding users to different roles

Creating a new user (Must know)

We are going to take a look at how to create a new user on AX 2012, and also look at how to use the import wizard form.

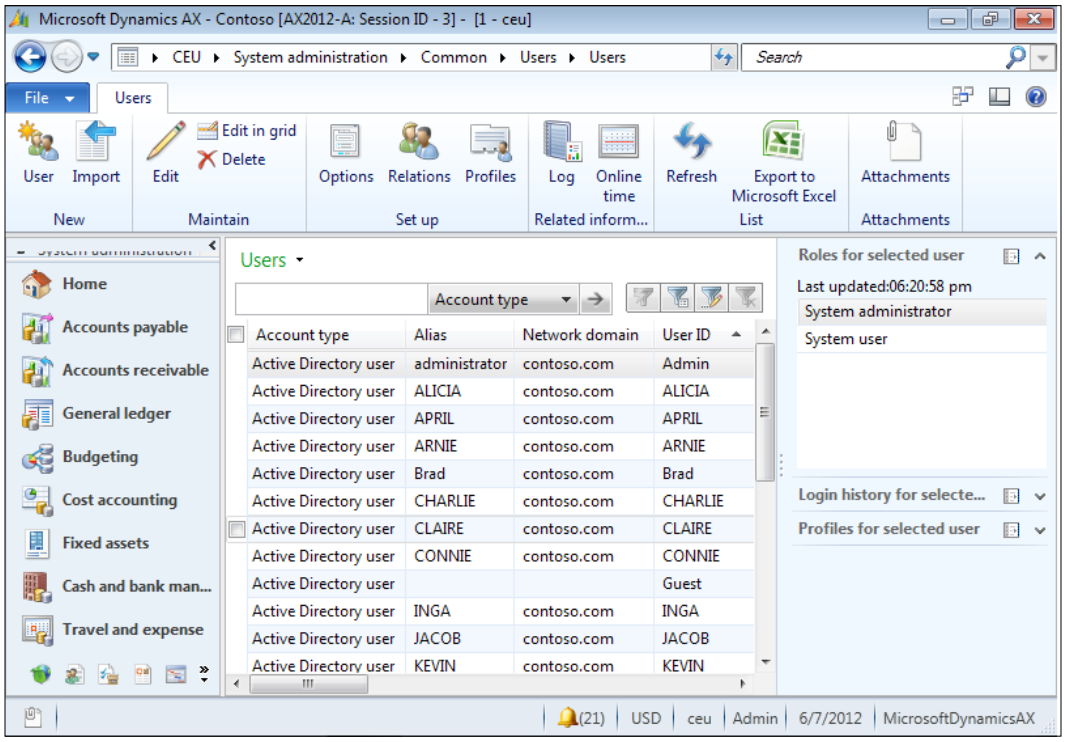
Getting ready

We have two methods to create users on Dynamics AX 2012. The first is to create a user in AX 2012 and map the user ID and username with the one created in Active Directory. The second method is to import the users created in the Active Directory to AX 2012, so you don't have to create users again in AX 2012. The latter method is the easier method, but I will explain how to perform both the methods.

How to do it...

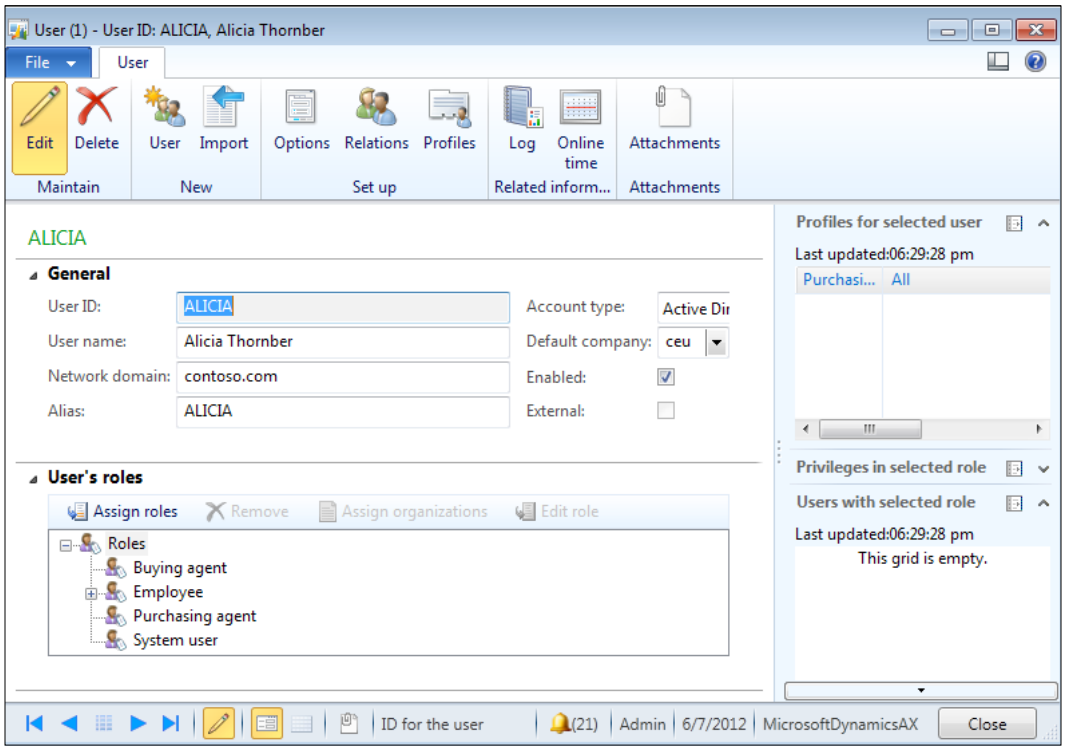
We are going to take a look at the necessary steps to create a new user with AX 2012:

1. In simple words, you have to know the user ID and username that was created before in Active Directory to create your new user in Dynamics AX.
2. The following screenshot shows the user form where you can view the list of all the users created before. You will also see, in the upper part of the window, a lot of functions that can be used by all users. The left-hand side of the window shows the roles, profiles, and privileges for every selected user.



3. Start AX Client and go to **System administration** | **Common** | **Users** | **Users**.

4. From the upper left-hand corner of the user list page, click on **User**.



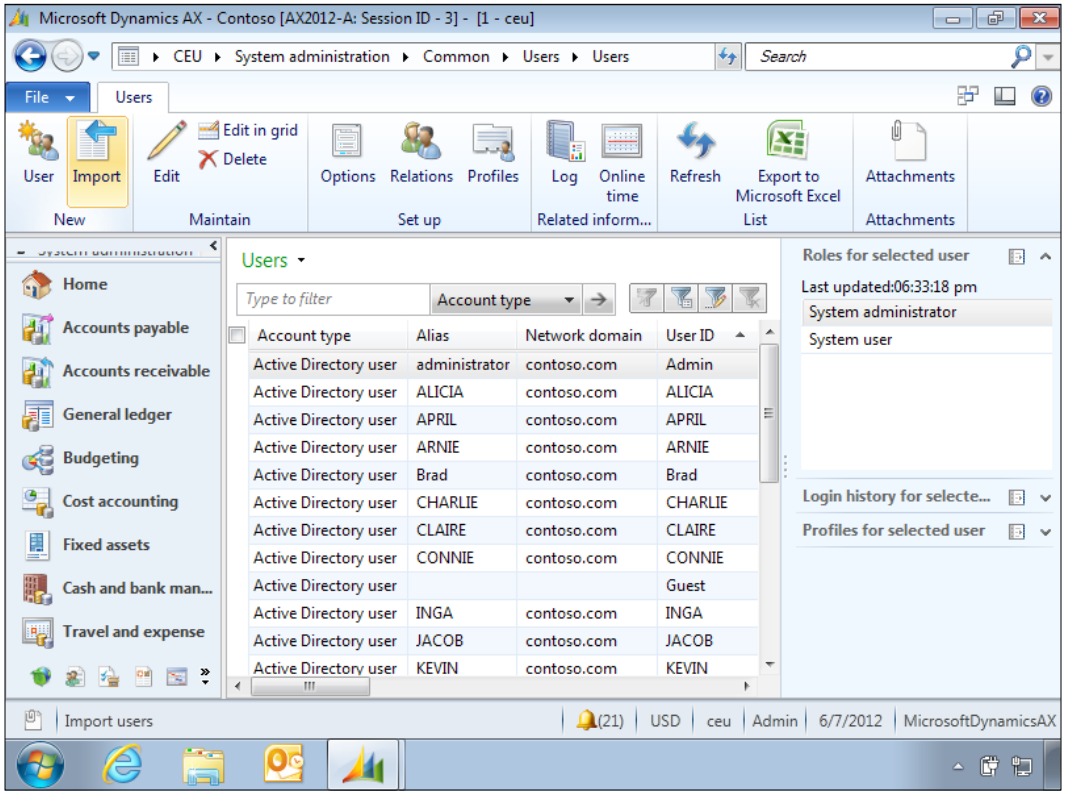
5. In the new window that appears, enter the user ID in the **User ID field**.
6. Enter the user's full name in the **User name** field.
7. In the **Network domain field**, enter the name of your network domain.
8. In the **Alias field**, enter either the name, a short form of the name, or the nickname of the user.
9. You have to confirm that the **Account type** field is set to **Active Directory User**.
10. Mark the **Enabled** checkbox.
11. Click on **Close** to close this window.

How it works...

These are the steps necessary to create a new user. After creating the user you can add or modify user-related information or add roles and privileges to the created user.

Importing users from Active Directory

From the user window that we saw in the previous screenshot, click on **Import** (as shown in the following screenshot) and start the **Active Directory Import Wizard**:

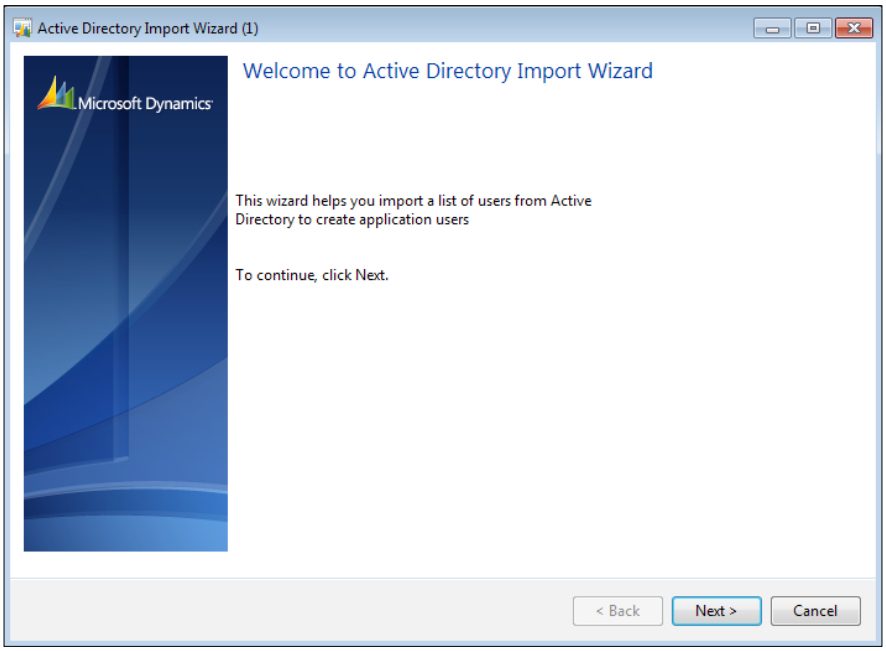


How to do it...

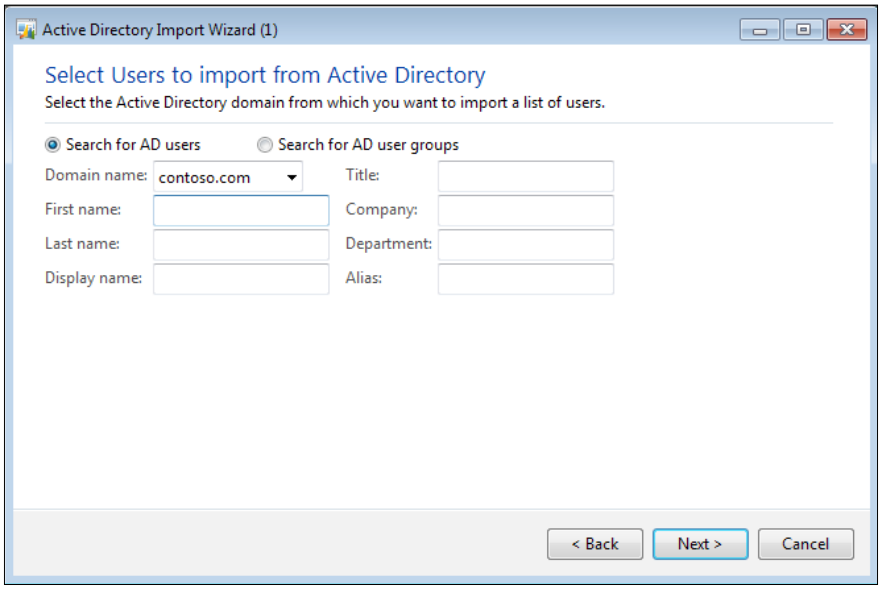
We are going to take a look at the steps necessary to import users from Active Directory:

1. Go to **System administration** | **Common** | **Users** | **Users**.

2. From the upper left-hand corner of the user list page, click on **Import**.



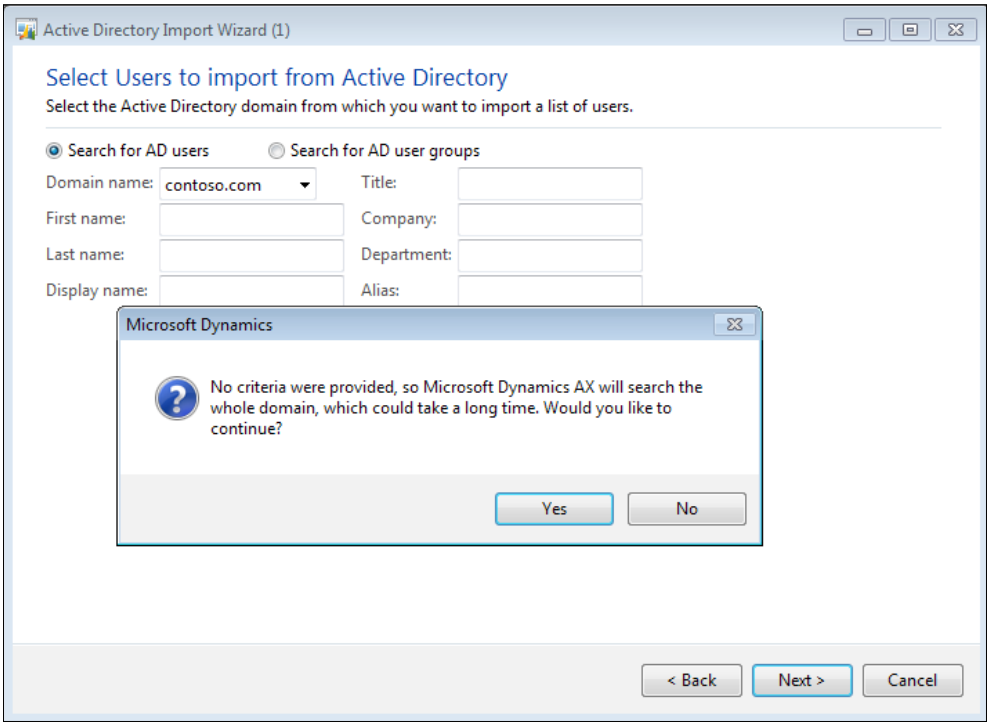
3. A new window called Active Directory Import Wizard appears.
4. Click on **Next**.



5. A window appears asking you whether to search for users or user groups from Active Directory. Select **Search for AD users**.

   You can determine a specific user to import by entering the **First name**, **Last name**, or **Display name** fields.

   Click on **Next**, if you want to select all users created in Active Directory.



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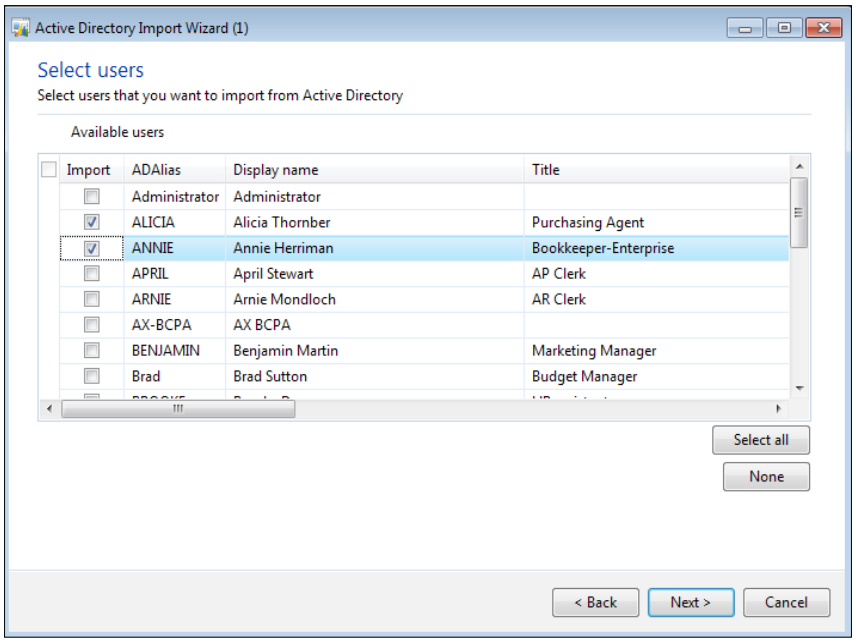
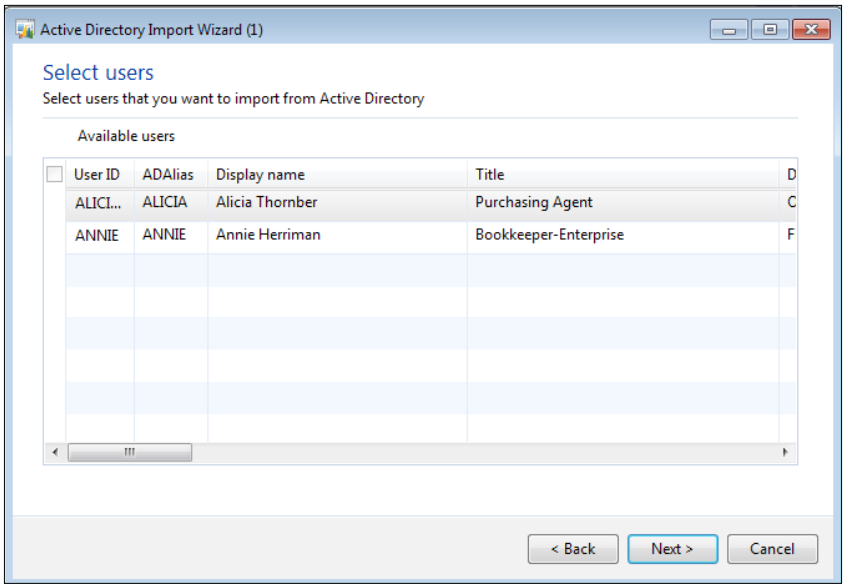
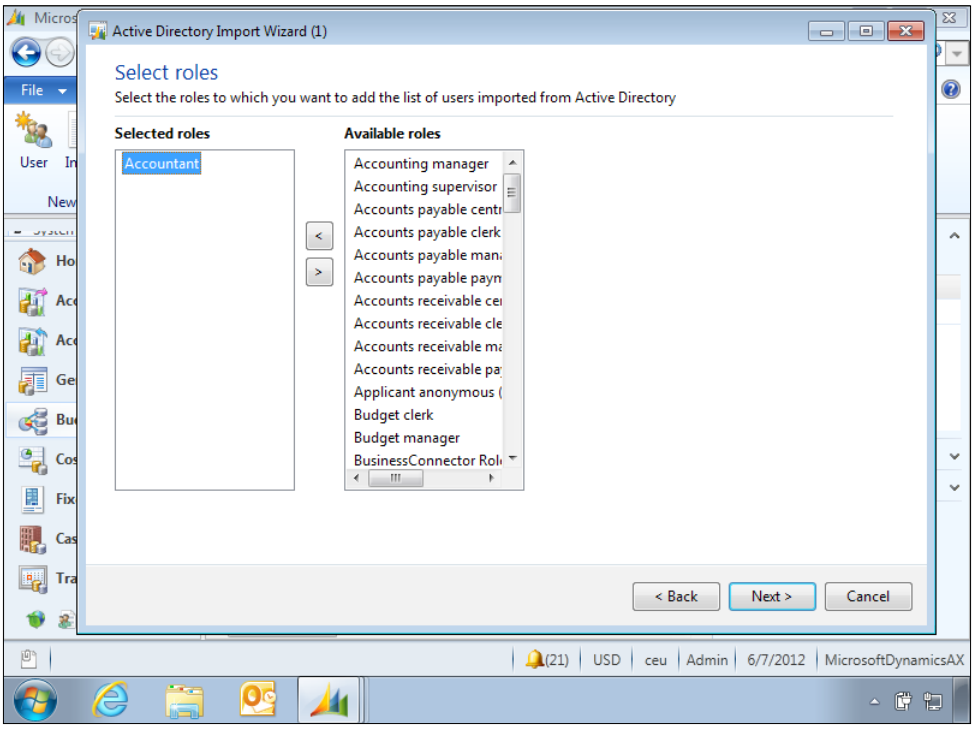
6. After you click on **Next**, a pop-up window will display a message, as shown in the previous screenshot. Click on **Yes** to proceed to the next step.



7. Select the users you want to import to the system by marking the checkbox beside each username, or click on **Select All** to select all users and then click on **Next**.

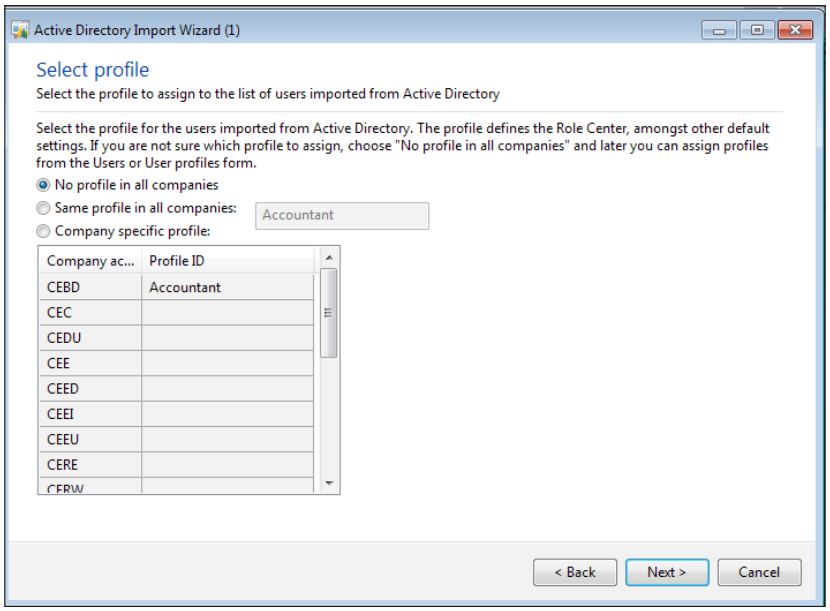


8. This window will display only those users that were selected in the previous step.

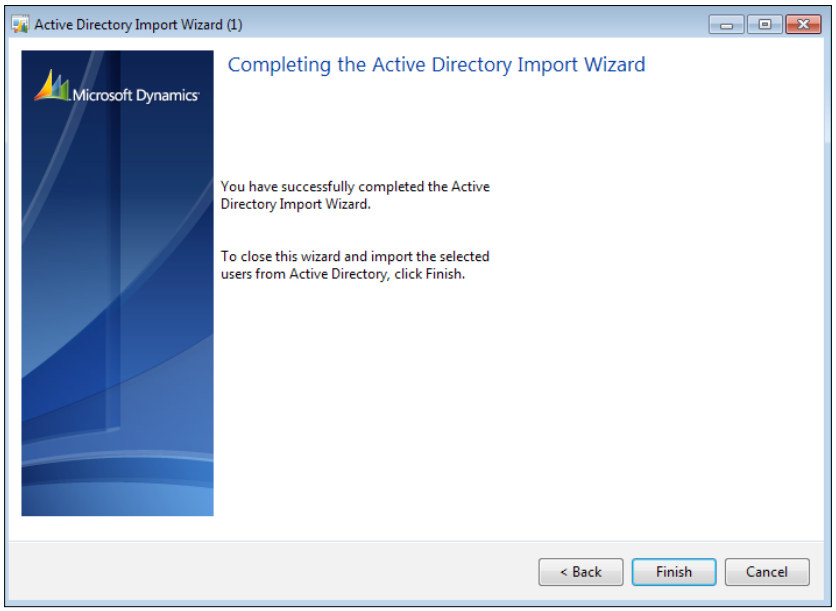


9. Select the role you want to add to the selected users by using the arrow buttons.

10. Select **No profile in all companies**. You can assign profiles from the user or user profile form later. Click on **Next**.



11. The next window will display that you have successfully imported the users from Active Directory. Click on **Finish** to close the window.



How it works...

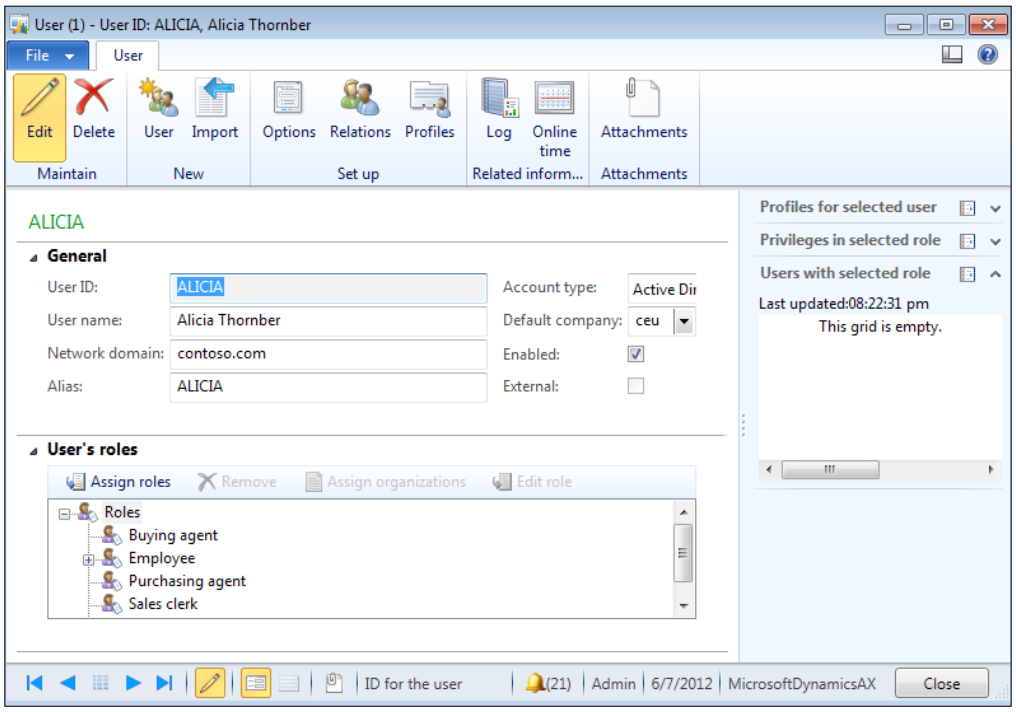
This is the easiest and simplest way to create your users. You just have to select the required users from Active Directory and make the necessary security settings; then follow the previously mentioned steps to import all users and user groups, and also assign predefined roles and user profiles to the imported users.

Creating a claim user (Should know)

In this section, we are going to take a look at how to create a new claim user.

How to do it...

The following screenshot shows the same user window we saw previously:



The process for creating a claim user is similar to the process we followed in the previous section, with a few changes in some steps; they are as follows:

1. Go to **System administration** | **Common** | **Users** | **Users**.
2. From the upper left-hand corner of the user list page, click on **User**.

3. In the new window that appears, enter the user ID in the **User ID field**.
4. Enter the user's full name in the **User name field**.
5. In the **Network domain** field, enter `FBAProvider`.
6. In the **Alias** field, enter either the name, a short form of the name, or the nickname of the user.
7. You have to confirm that the **Account type** field is set to **Claims user**.
8. Mark the **Enabled** checkbox.
9. Click on **Close** to close this window.

How it works...

We performed the same steps we did when creating a new user. We just changed the account type for the selected user to differentiate the organization's Active Directory users from any other user that doesn't belong to the organization.

Users created so far don't have any permissions. We will see how to create permissions in the later recipes.

Claim users: These are users who are not in an organization's Active Directory, like vendors or customers who need to access Enterprise Portal.

Setting up user (Should know)

In this section, we are going to take a look at how to complete the user's setup.

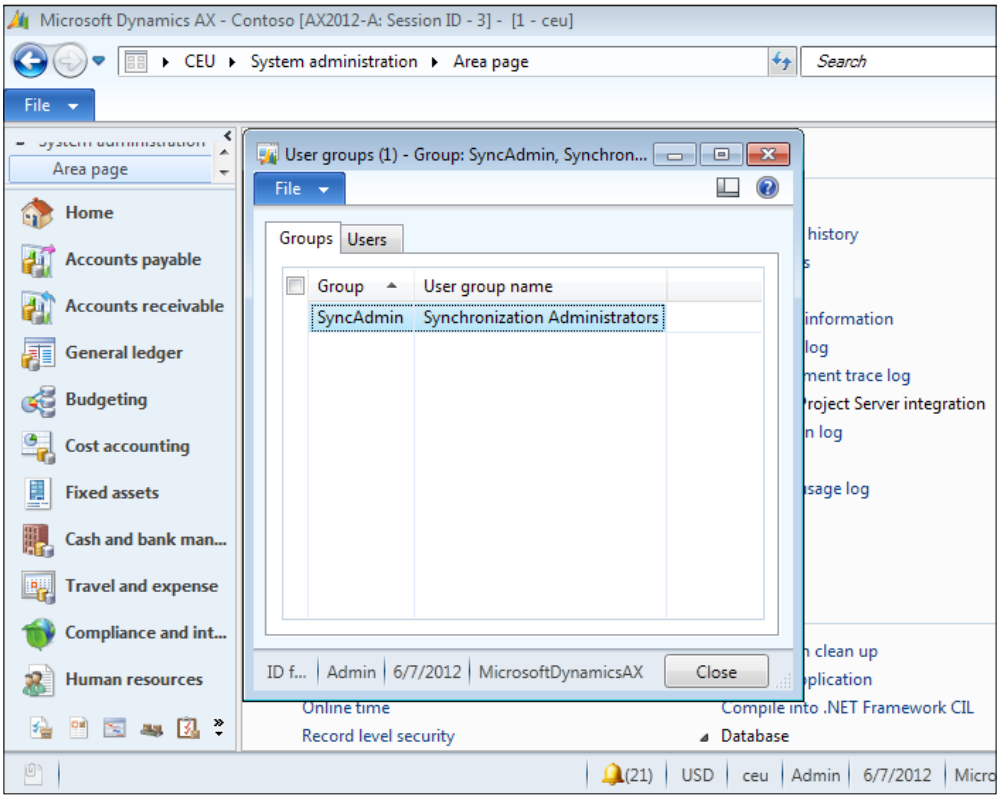
Getting ready

After creating a new user, whether created on AX 2012 or imported from active directory, you have to configure the setup that is needed for every user. The steps are as follows:

- Assign user to user group
- Set up user options
- Set up user relations
- Set up user profile

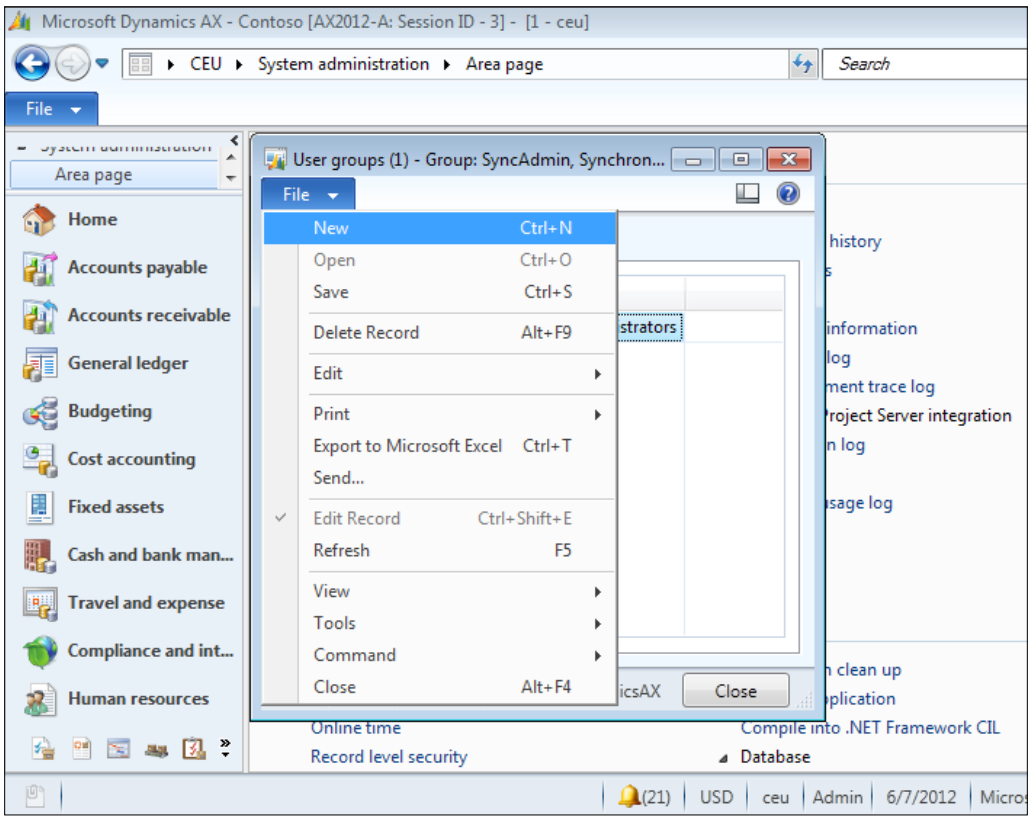
How to do it...

The following screenshot shows the user group form that has two tabs—one for user groups and the other to assign all users listed to the selected group. Also notice the file drop-down menu that enables you to save, edit, and print your work in this form.

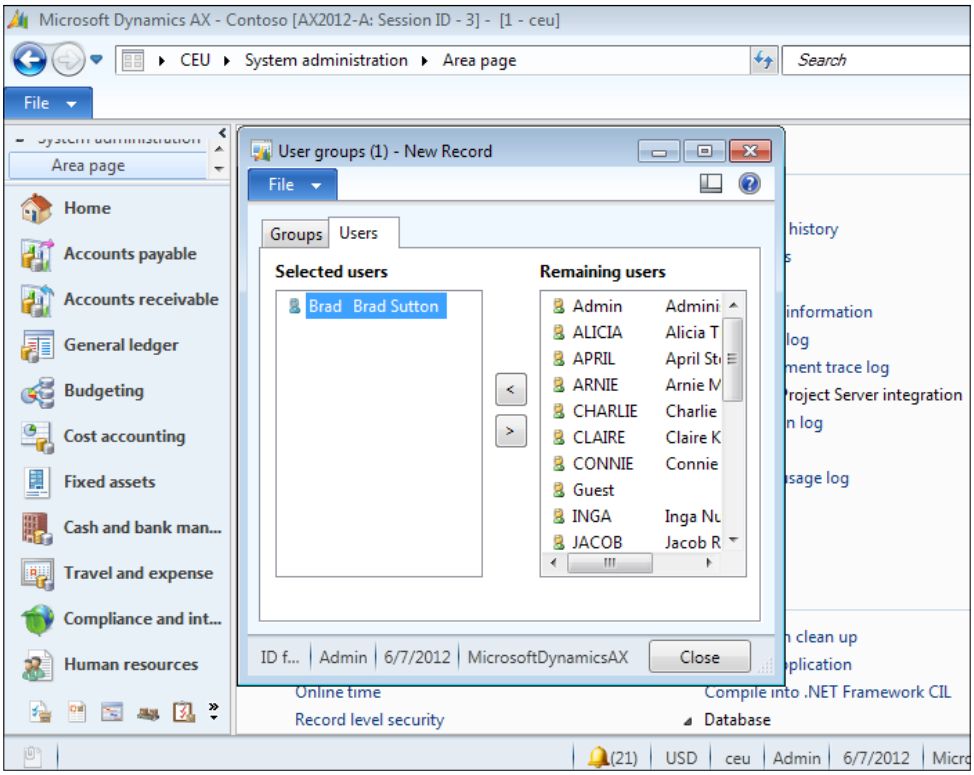


We are going to take a look at the steps necessary to set up user groups, options, relations, and profiles:

1. Go to **System administration** | **Common** | **Users** | **User Group**.



2. Click on **File** and select **New**, or press *Ctrl* + *N*, to create new user groups.
3. Enter the group ID in the **Group ID field**.
4. Enter the group name in the **User group name** field.
5. Click on **Save**, or press *Ctrl* + *N*, to save your record.



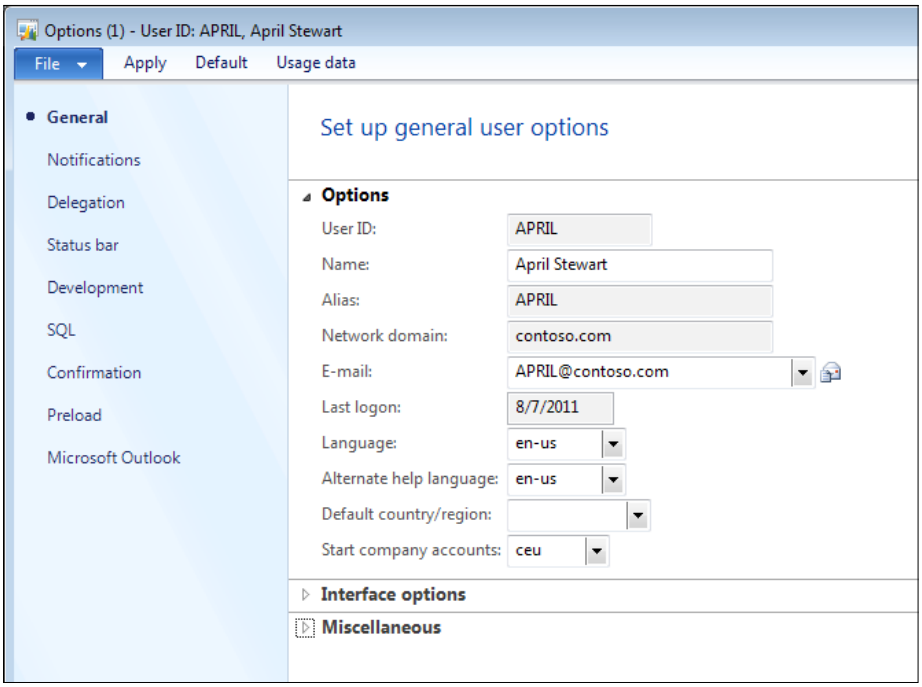
6. Switch to the **Users tab**.
7. Select the users related to the created group from the **Remaining users** table using the arrows at the center of the window.
8. Click on **Close** to close the window.

How it works...

In this form you can create a user group or assign users to user groups by selecting the appropriate group, and move to the users tab to view the list of users created before. By clicking on the upper arrow in the middle of the screen, you can move the user from the **Remaining user** table (right-hand side column) to the **Selected users** table (left-hand side column). To remove a user from the selected group, just click on the lower arrow in the middle of the screen and you will notice the movement of the selected user from the left-hand side column to the right-hand side column.

Setting up user options (Should know)

The following screenshot illustrates the general user options and the basic and simple information that needs to be filled. To open this window, click on the **Option** button on the user's main form.



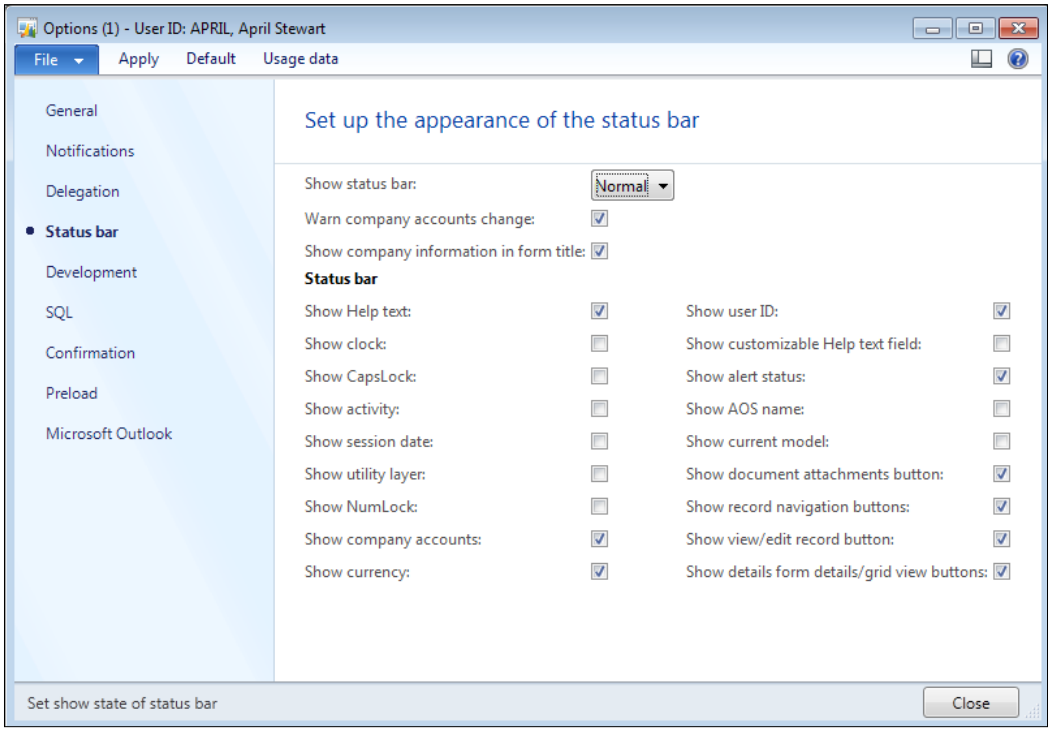
How to do it...

1. Go to **System administration** | **Common** | **Users** | **Users**.
2. From the upper part of the user list page, click on **Options.**
3. The following user options are available:

   The **General** tab:

   - You will notice that the important information of the user has been displayed, like **User ID**, **Name**, **Alias**, and **Network domain**
   - Enter the e-mail address of this user in the **E-mail** field
   - You can find out the last login date of the user to the system from the **Last logon** field
   - From the **Language** field you can select the required language for the user who will deal with the whole system

- From the **Start company accounts** field you can select the default starting company that will be displayed to the user after running the AX client.



The **Status bar** tab:

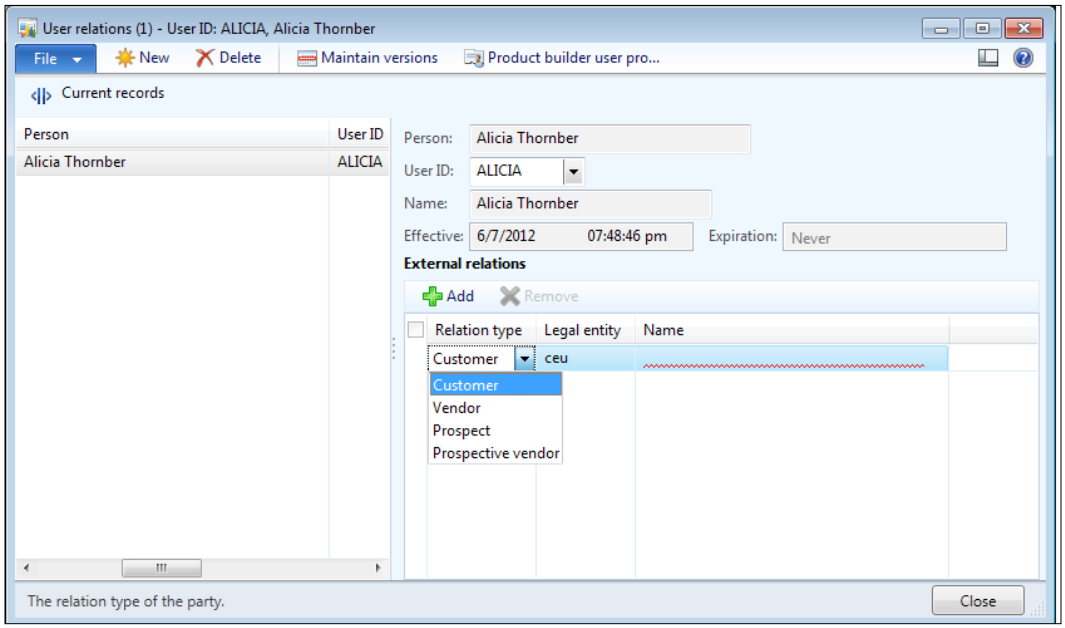
- You can mark or unmark the checkboxes displayed, according to the user's preferences. This information will be shown on the status bar.
- The **Warn company accounts change** checkbox will, if marked, give you a warning message when moving through the companies in the system.

How it works...

It is just a few steps to add the user's options and also a way of personalizing your AX home page, viewing your AX interface, dealing with notifications, and synchronizing your Outlook e-mail account with AX 2012. Feel free to personalize your AX!

Setting up user relations (Should know)

The following screenshot shows how to determine the user relation with the organization, that is users that have access to the business portal, like customer, vendor, or prospect. Just add the user relation type from the list:



How to do it...

We are going to take a look at the steps necessary to set up user relations:

1. Go to **System administration** | **Common** | **Users** | **Users**.
2. Select the required user from the user's list.
3. From the upper part of the user list page, click on **Relations**.
4. Click on **Add** at the centre of the displayed window to add a relation to the selected user.
5. A new record line is created. Select the relation type from the drop-down list (**Customer**, **Vendor**, **Prospect**, and **Prospective vendor**).
6. Type a name for the selected relation.
7. Click on **Close** to close the window.

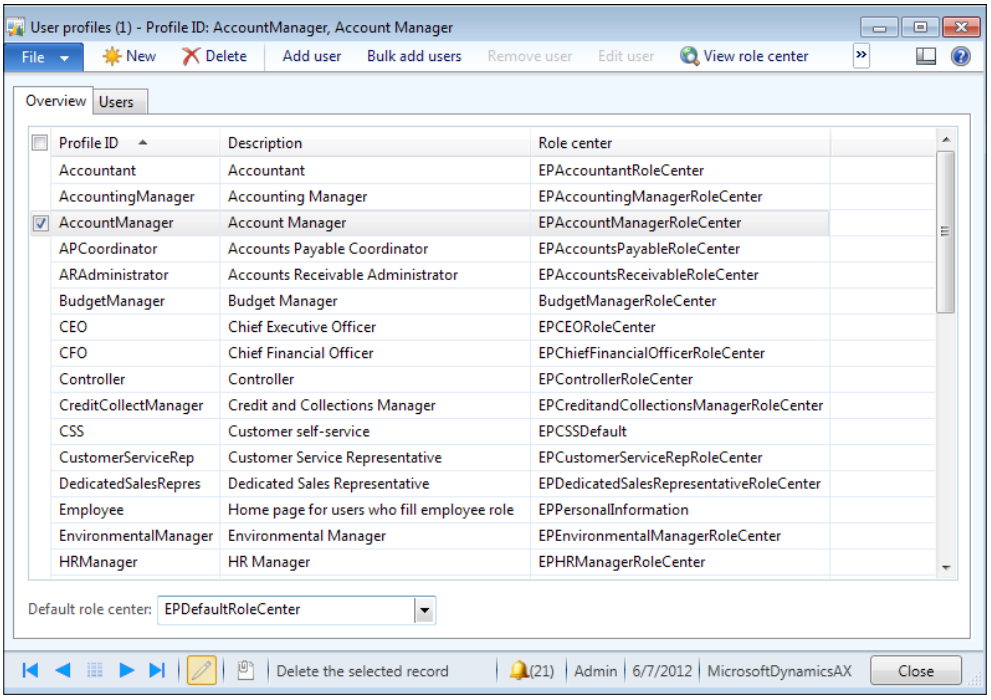
How it works...

Set up the user relation by selecting the appropriate relation from the drop-down list, which consists of **Customer**, **Vendor**, **Prospect**, **Prospective Vendor**, and also which company/companies he/she can access. This option can be made for claim users who can access Enterprise Portal.

Adding a user to a predefined profile (Should know)

We are going to take a look at how to associate a user to a predefined profile user, and also the companies that will be available for the selected profile.

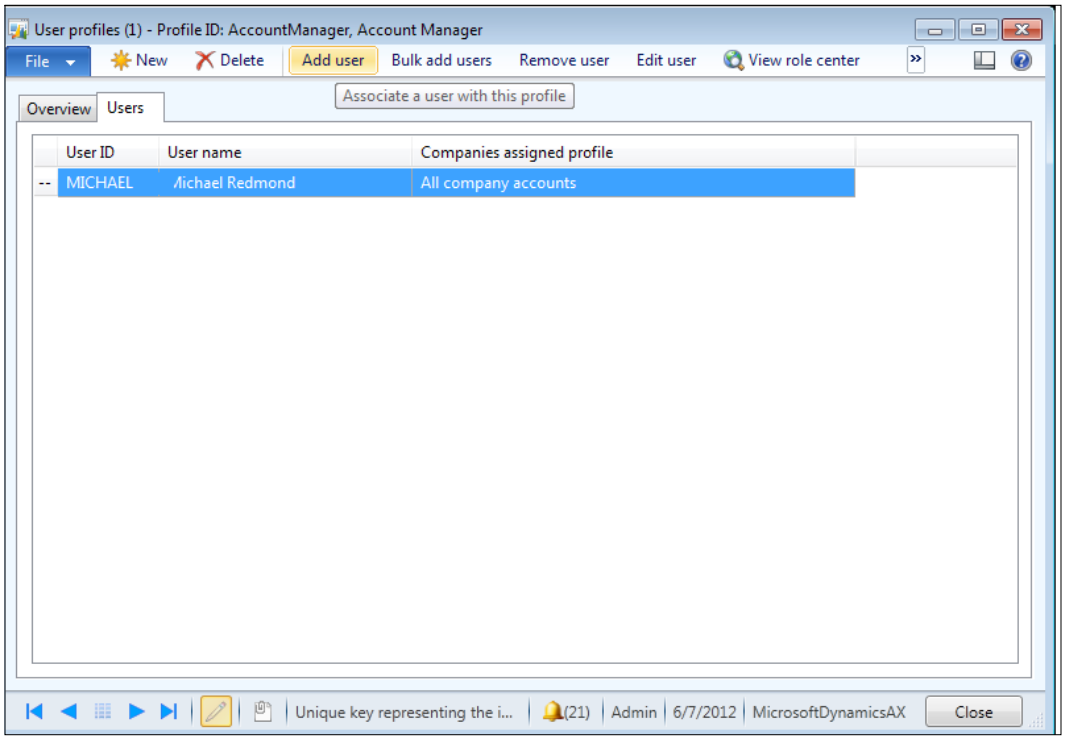
The following screenshot shows predefined profiles listed on the **Overview** tab. Select the desired profile and switch to the **Users** tab to add users to the selected profile.



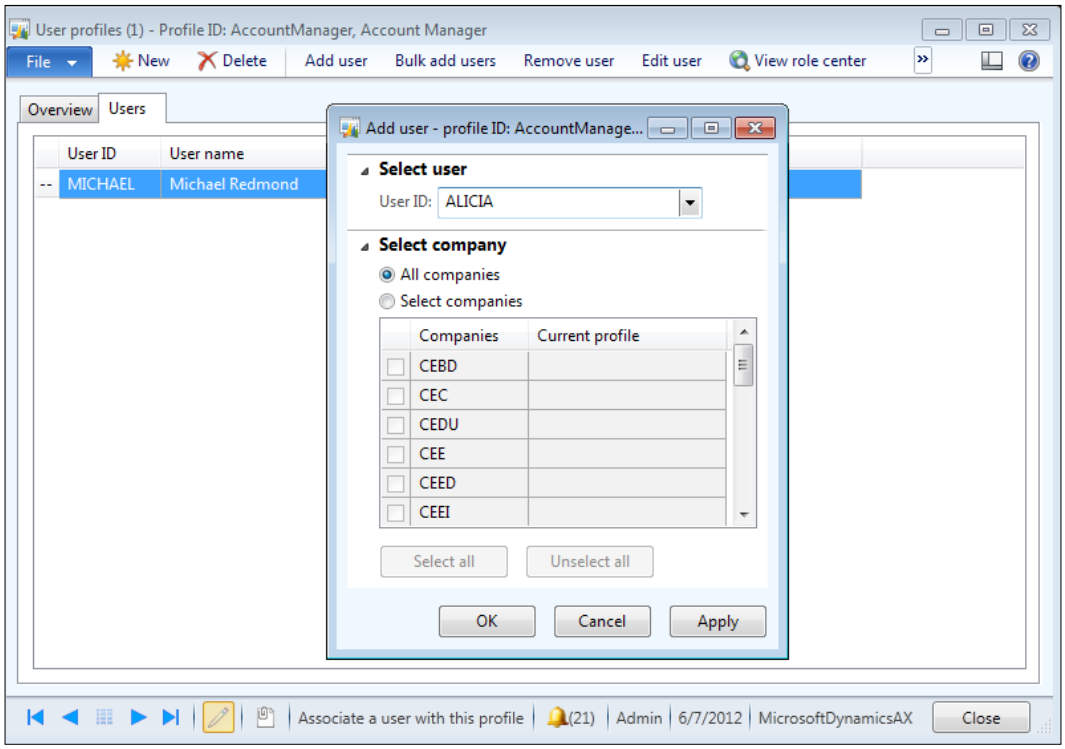
How to do it...

You will notice the upper bar on the window that displays the buttons to add, remove, or edit users. Perform the following steps to add a user to a predefined profile:

1. Go to **System administration** | **Common** | **Users** | **Users**.
2. Select the required user from the user's list.
3. From the upper part of the user list page, click on **Profile**.
4. Mark the appropriate profile ID displayed in the list.
5. Switch to the **Users tab**.



6. Click on **Add user** to associate a user with the profile selected.



7. From the window that appears, select the user from the **User ID** drop down list.
8. Select whether the user will have access to all companies or specific companies only.
9. Click on **OK** to apply the changes you made in the window in the previous step.
10. A new record line will be created in the grid table with the assigned user.
11. Click on **Close** to close the window.

How it works...

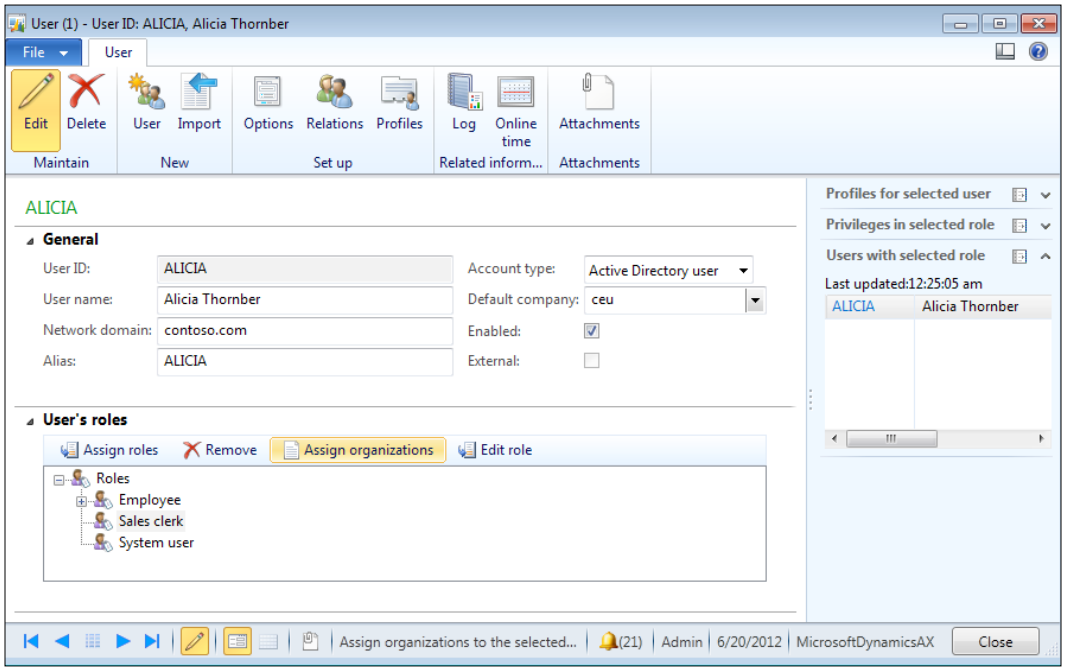
These steps show how to assign a user to a predefined user profile, and also determine the companies that will be available for the selected profile.

You can also add a bulk of users, view the role centre related to every profile, or edit the users themselves by selecting the appropriate button from the upper part of the window.

Granting a user access to a specific company (Must know)

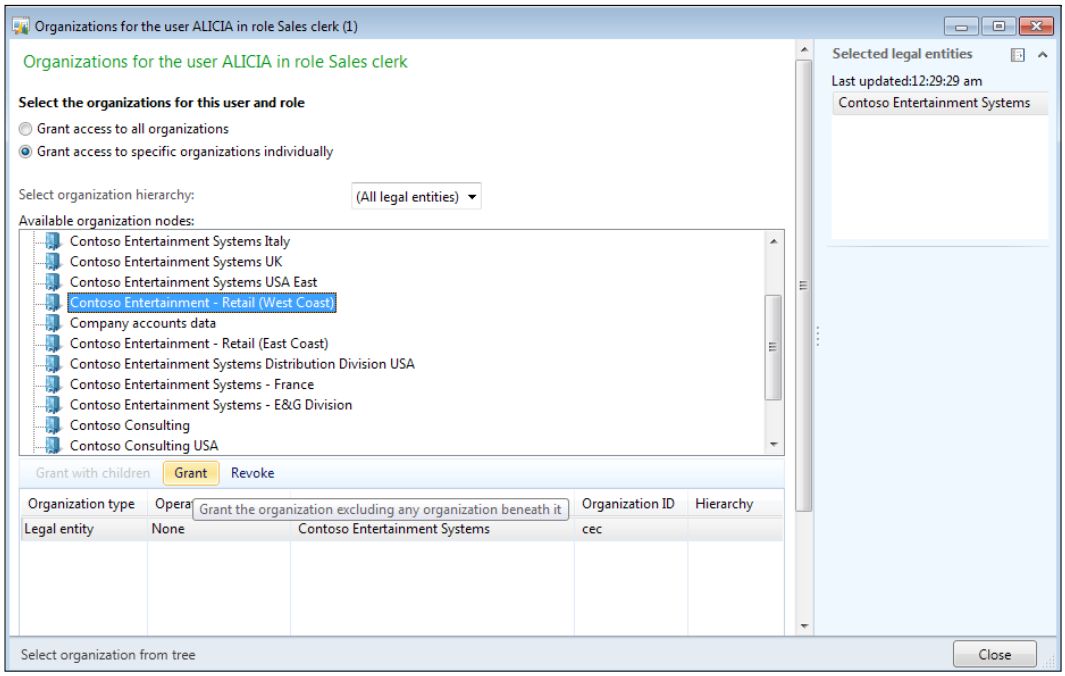
We are going to take a look at how to assign users to access certain specific companies.

The following screenshot shows user information after it is selected. In the lower part of the window you will notice the **Assign organizations** button; this is where you will assign the user access to specific companies or all companies.



How to do it...

1. Go to **System administration** | **Common** | **Users** | **Users**.
2. Select the required user from the user list.
3. Click on the **Assign organizations** button at the centre of the form.



4. Select **Grant Access to specific organization individually**.
5. Set **Select organization hierarchy** to **(All legal entities)** to view all the companies available.
6. Select the company you want the user to access and click on **Grant** to grant this user permission to access the selected company.
7. To grant user access for multiple companies, repeat the previous two steps.
8. You can click on **Revoke** to unassign the user access to the selected company.
9. After finishing, click on **Close** to close the form.

How it works...

The steps in the previous section show how to assign a user access to a specific company using the **Grant** button, and also how to unassign a specific user's access to a specific company using the **Revoke** button. You need to be certain when you select the required company from the available organization nodes, or you may grant user access to all organizations available on the system.

Tracking user-related information (Must know)

In this section, we are going to track the user's login and logoff records through the user-related information form.

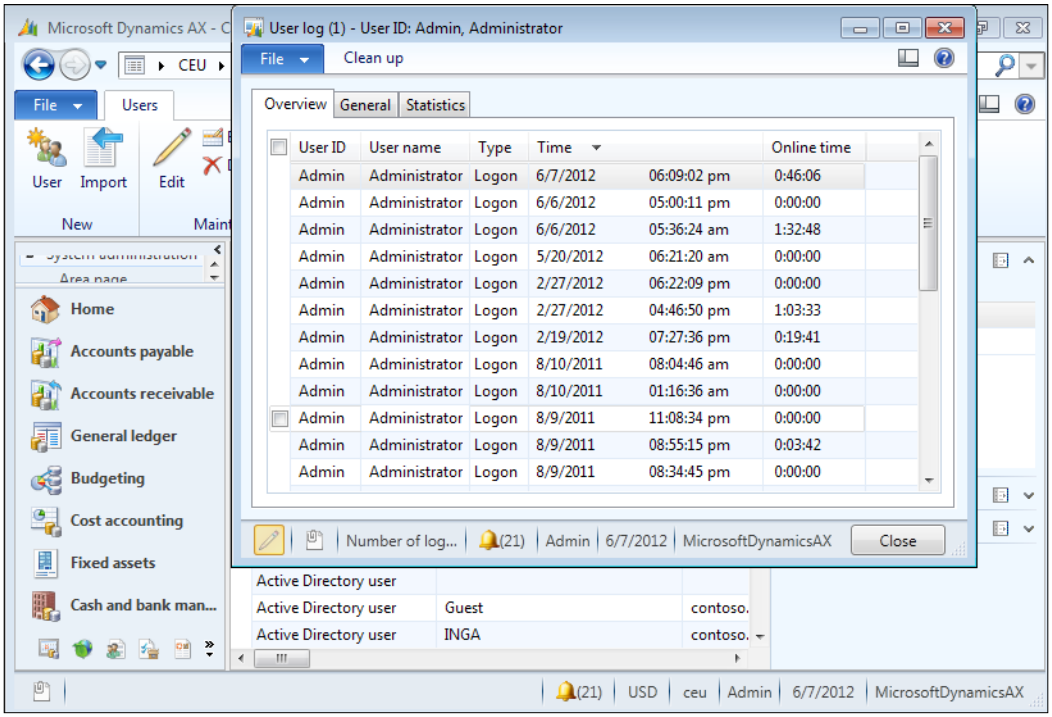
Getting ready

In Microsoft Dynamics 2012 AX, we can track user-related information such as the user's login and logoff date, and time or the created date and time by using the user's log and online time report.

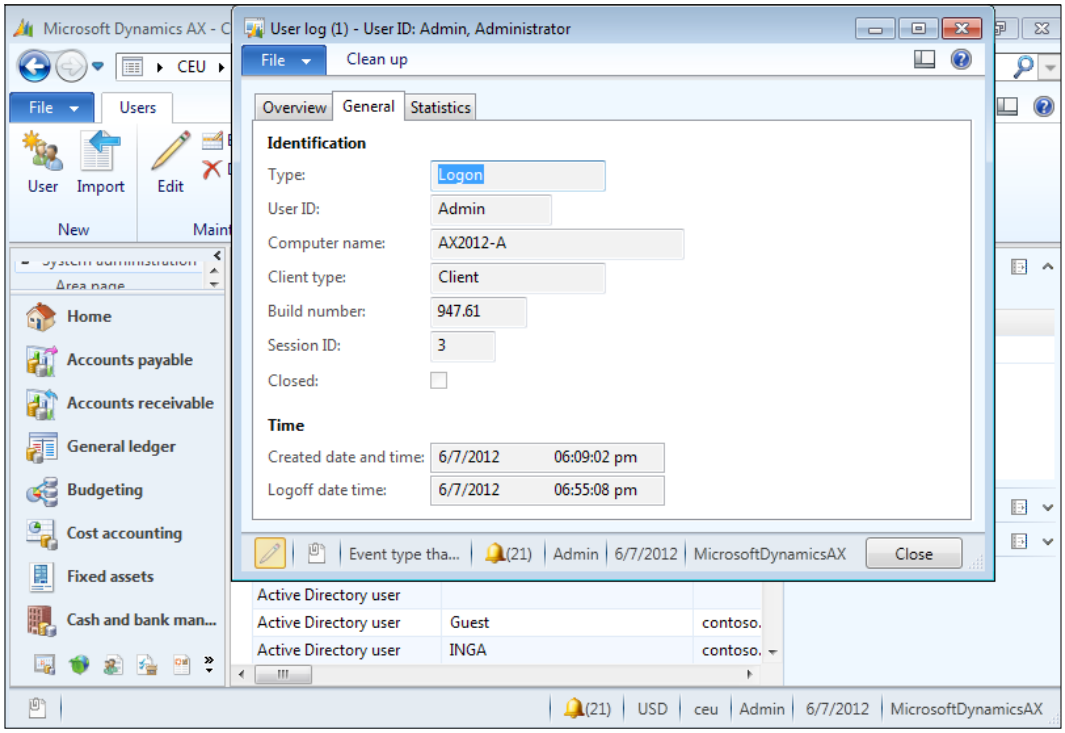
How to do it...

We are going to take a look at the steps necessary to view the user's log and their online time:

1. The following screenshot shows the online time and the logon/logoff date for all listed users. After selecting the required user, you can move to the other tabs, **General** tab and **Statistics** tab, to view detailed information.



2. Go to **System administration** | **Common** | **Users** | **Users**.
3. Select the required user from the list of users.
4. From the upper part of the user list page, click on **Log.**
5. A window appears that shows a grid table with the logon date and time for the selected user.
6. Select the line record from the grid table.



7. Switch to the **General** tab to view detailed information for any record.
   - **User ID**
   - **Computer name**
   - **Date and time of creation**
   - **Logoff date and time**
8. Click on **Close** to close the window.

How it works...

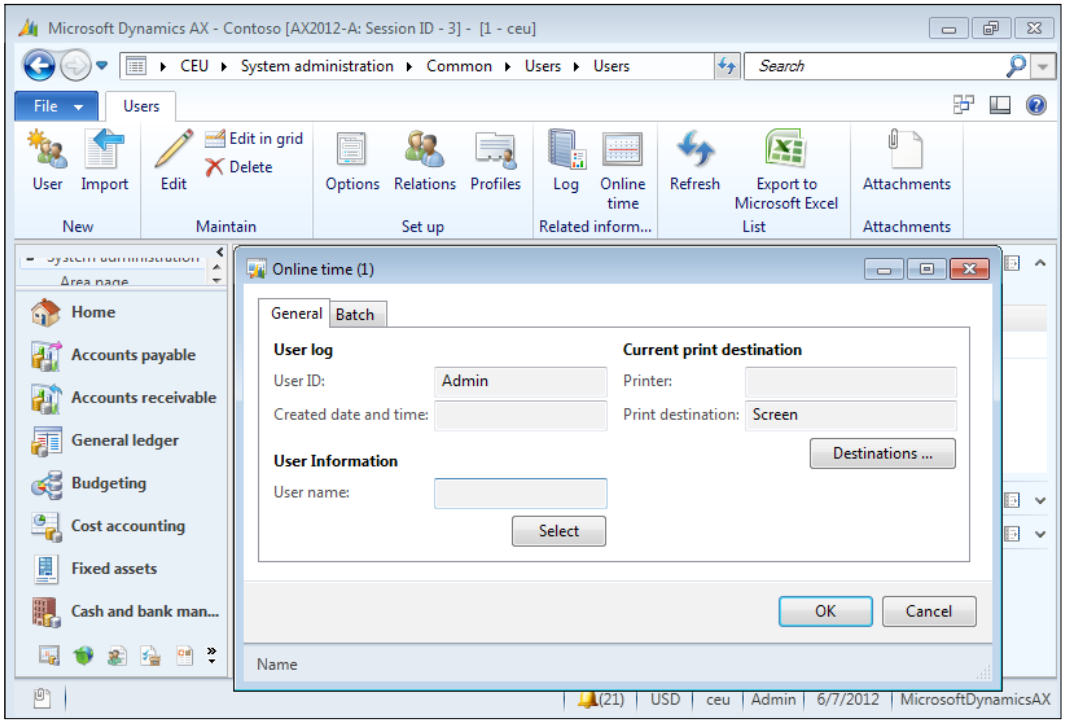
Every time users log on or log off Dynamics AX 2012, this log will be filled automatically with the date and time spent. You can clean up this log by clicking on the **Clean up** button located on the upper bar of the window.

There's more...

After being oriented with the user's log, you can also print detailed information by viewing the user's online time; just determine your print destination.

Viewing the user's online time

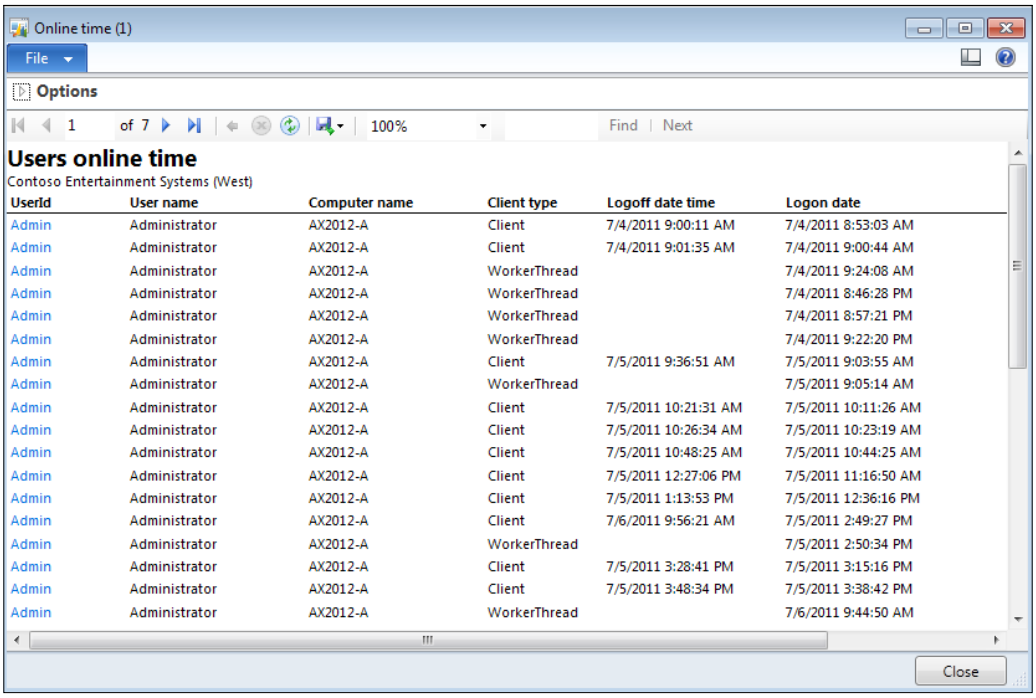
The following screenshot shows a way to view and print the user logon/logoff time and date and time spent using the system.



The steps to view the user's logon/logoff data are as follows:

1. Go to **System administration** | **Common** | **Users** | **Users**.
2. Select the required user from the list of users.
3. From the upper part of the user list page, click on **Online Time.**

4. A new window appears; click on **Select** to select a user.
5. Click on **OK** to view the report.



UserId	User name	Computer name	Client type	Logoff date time	Logon date
Admin	Administrator	AX2012-A	Client	7/4/2011 9:00:11 AM	7/4/2011 8:53:03 AM
Admin	Administrator	AX2012-A	Client	7/4/2011 9:01:35 AM	7/4/2011 9:00:44 AM
Admin	Administrator	AX2012-A	WorkerThread		7/4/2011 9:24:08 AM
Admin	Administrator	AX2012-A	WorkerThread		7/4/2011 8:46:28 PM
Admin	Administrator	AX2012-A	WorkerThread		7/4/2011 8:57:21 PM
Admin	Administrator	AX2012-A	WorkerThread		7/4/2011 9:22:20 PM
Admin	Administrator	AX2012-A	Client	7/5/2011 9:36:51 AM	7/5/2011 9:03:55 AM
Admin	Administrator	AX2012-A	WorkerThread		7/5/2011 9:05:14 AM
Admin	Administrator	AX2012-A	Client	7/5/2011 10:21:31 AM	7/5/2011 10:11:26 AM
Admin	Administrator	AX2012-A	Client	7/5/2011 10:26:34 AM	7/5/2011 10:23:19 AM
Admin	Administrator	AX2012-A	Client	7/5/2011 10:48:25 AM	7/5/2011 10:44:25 AM
Admin	Administrator	AX2012-A	Client	7/5/2011 12:27:06 PM	7/5/2011 11:16:50 AM
Admin	Administrator	AX2012-A	Client	7/5/2011 1:13:53 PM	7/5/2011 12:36:16 PM
Admin	Administrator	AX2012-A	Client	7/6/2011 9:56:21 AM	7/5/2011 2:49:27 PM
Admin	Administrator	AX2012-A	WorkerThread		7/5/2011 2:50:34 PM
Admin	Administrator	AX2012-A	Client	7/5/2011 3:28:41 PM	7/5/2011 3:15:16 PM
Admin	Administrator	AX2012-A	Client	7/5/2011 3:48:34 PM	7/5/2011 3:38:42 PM
Admin	Administrator	AX2012-A	WorkerThread		7/6/2011 9:44:50 AM

6. You can print the user online time, save it as a file by selecting the **Print** option, or press *Ctrl + N.*
7. Click on **Close** to close the window.

This is just a simple way to print and view the user logon/logoff information in a report format; this can be done by selecting the required user and, from the window that appears, selecting your printer and print destination.

Assigning users to different roles (Must know)

We are going to take a look at how to assign users to security roles.

Getting ready

Roles can be assigned to users in two ways: adding roles to users and adding users to roles. This will be shown in different windows and forms.

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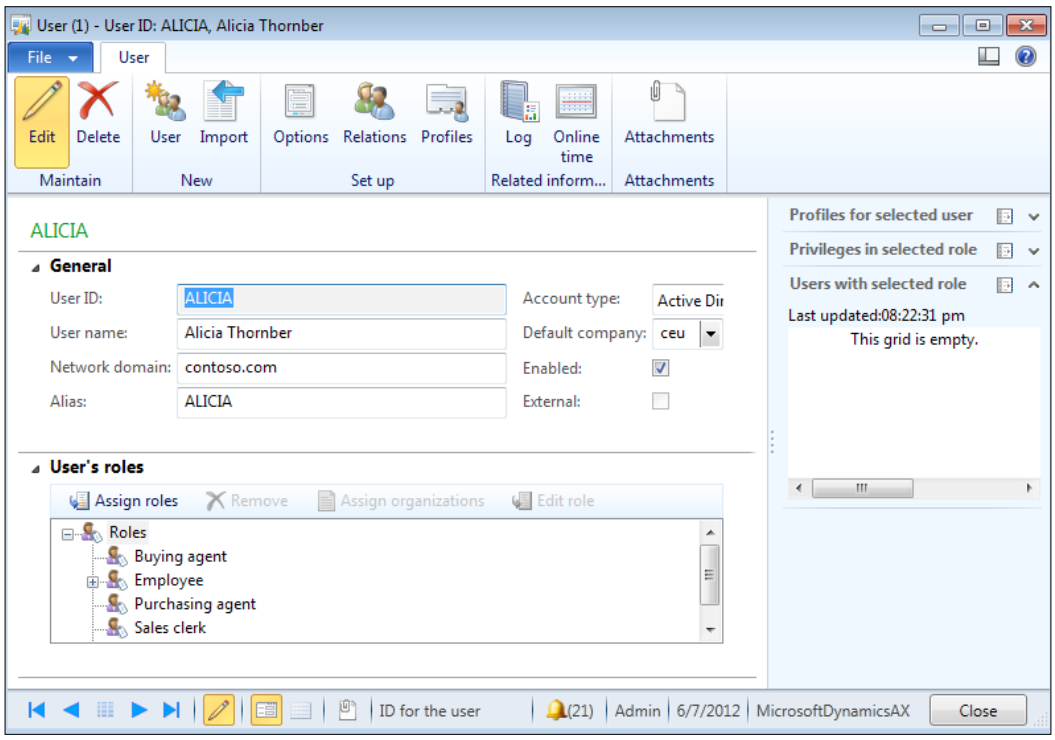
Roles are groups or duties for a job function that represent the behavior of a person that plays a particular role, including a defined set of application access privileges.

How to do it...

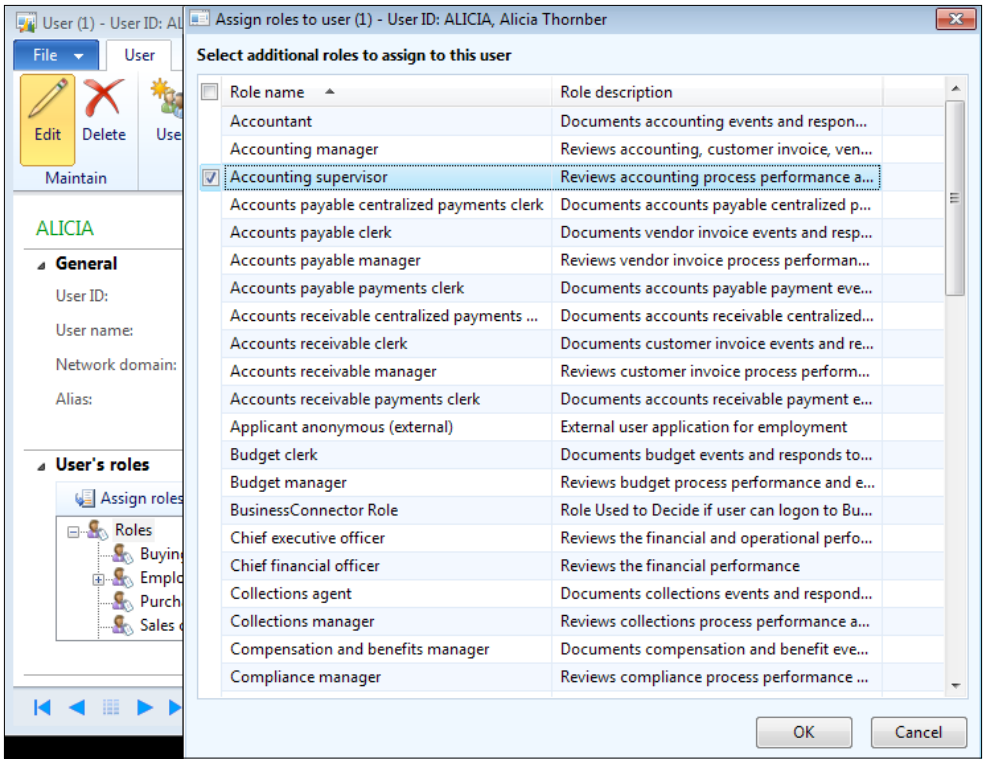
We are going to look at the steps needed to add roles to users:

1. The following screenshot shows the user form once the required user is selected. At the lower part of the window, click on the **Assign roles** button to add a role to the selected user.



2. Go to **System administration** | **Common** | **Users** | **Users**.
3. Select the required user from the user's list.
4. From the upper left-hand corner of the user list page, click on **Edit**.
5. In the centre of the user form, click on **Assign roles**.

   The following screenshot appears after clicking on **Assign roles**. Select a role from the list of predefined roles.



6. Select an option in the **Role name** menu to assign to the selected user.
7. Click on **OK**.
8. Click on **Close** to close the user form.

How it works...

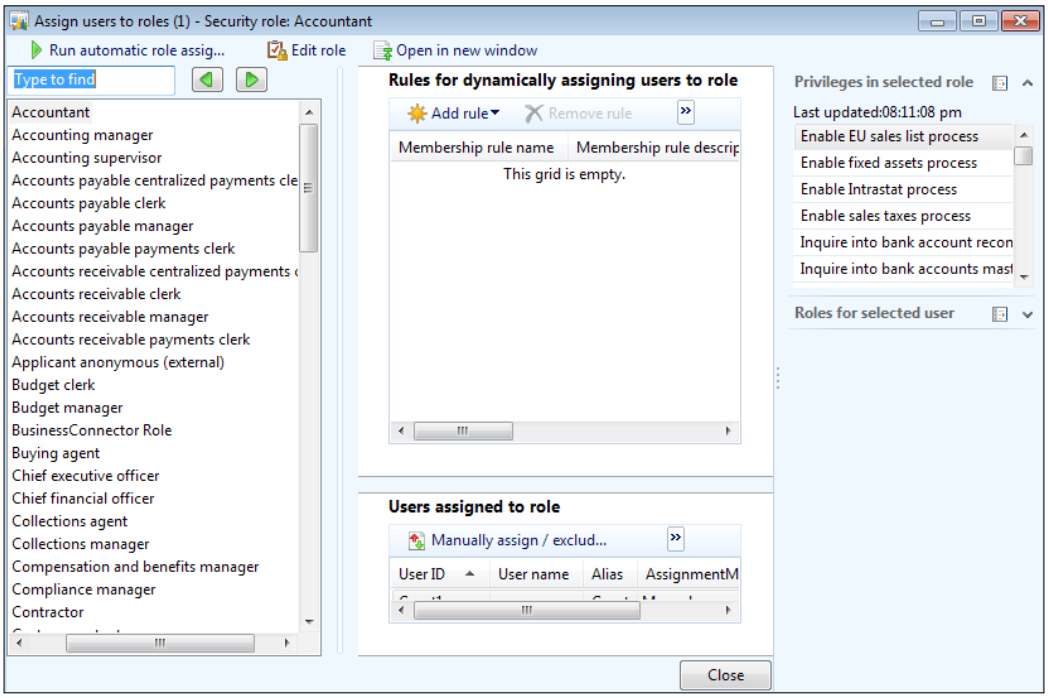
We selected the user first, and then assigned a role from the predefined roles list by marking the role from the checkbox that appears for every record.

Adding users to the different roles (Should know)

In this section, we are going to look at the method to add users to security roles.

The following screenshot shows the **Assign users to roles** form. The left-hand side of the form contains a list of predefined roles and gives you the ability to find a specific role to select your role easily.

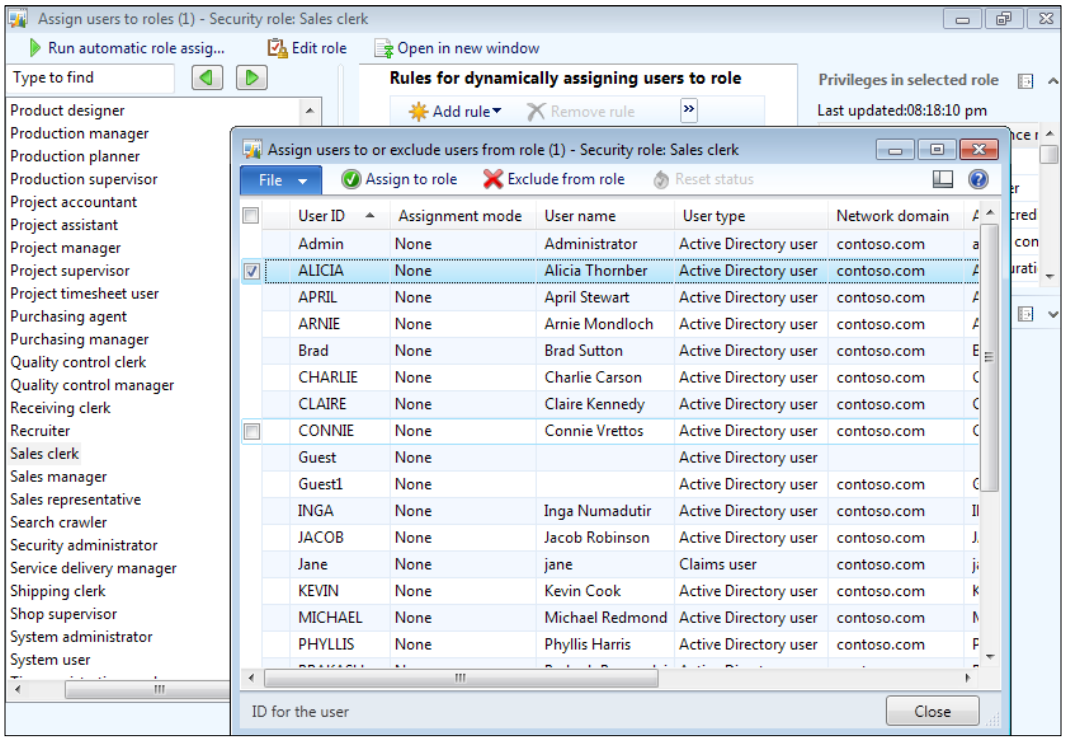
At the center of the window you will find the **Manually assign / exclude users** button. This button is used to add a user to the selected role.



How to do it...

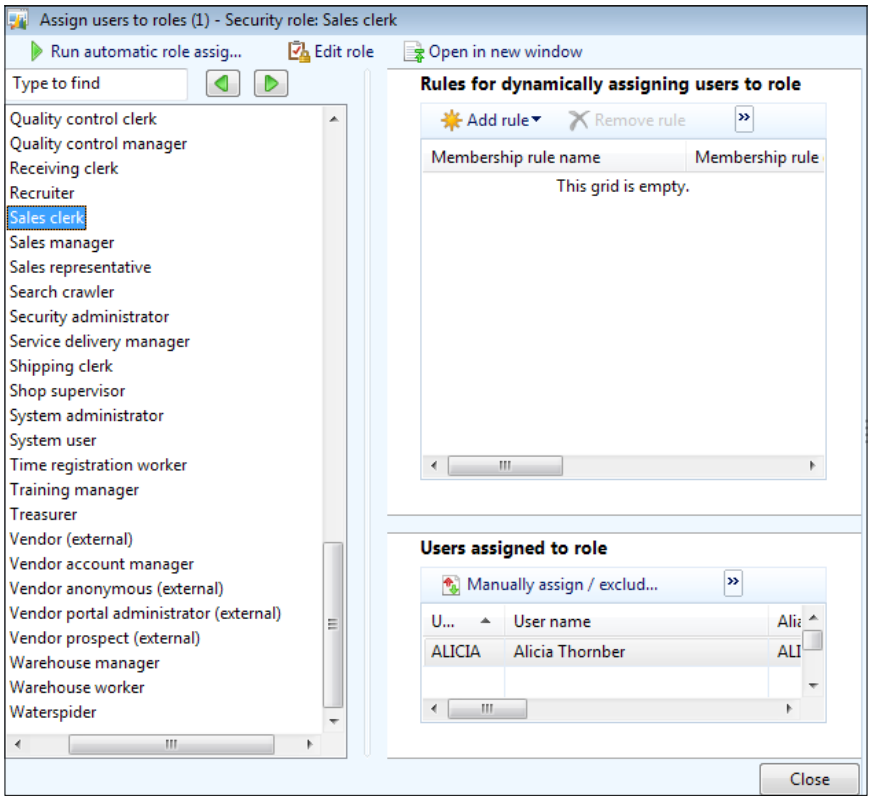
1. Go to **System administration** | **Setup** | **Security** | **Assign users to roles**.
2. Select a role from the list on the left-hand side of the window.
3. Click on **Manually assign/exclude users**, located at the centre of the form.

4. You can select the user you want to assign to the selected role by clicking on the **Assign to role** button at the top of the window, as shown in the following screenshot:



5. From the window that appears, select the required user from the list.
6. You can select more than one user by clicking on **Mark all users**.
7. Click on the **Assign to role** button at the top of the form.
8. You will notice that the marked line record changes to a green color.
9. Click on **Close** to close the window.
10. You will notice that the selected user has been added to the role of **Sales clerk**, selected in the main form.

11. Click on **Close** to close the window.



How it works...

By using this method, we added a user to the security role. We used the **Assign users to roles** form, selected the role from the left-hand side of the window, and then assigned a user to the selected role. You can use the same steps to exclude a user from a specific role by using the **Manually assign / exclude users** button. For editing a specific role, just select the role first and then click on the **Edit role** button on the upper bar of the same form.

In this section, we will learn how to assign roles based on their responsibilities and business processes, and also make modifications to predefined roles.

Simply, security roles are defined all at once, for all organizations and all users must be assigned to at least one security role in order to have access to Microsoft Dynamics AX, so it is very simple and also saves time because you don't have to manage the access for every user separately.

Besides the simplicity of the security role, they can be organized into a hierarchy, so you can define a security role based on another role. It is like the relationship between a parent and a child, but when talking about roles, you can have one or more child or one or more parent role, and all privileges and duties get inherited from the parent role to the child role.

Adding a security role (Become an expert)

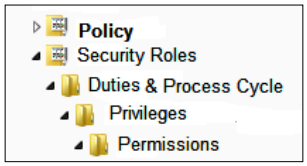
In this section, we will learn the concept of security roles and how to add security role.

Getting ready

Before getting involved with the steps required to add the security role, we need to understand the concept of the security role itself and other, related security terms.

We have to consider that every organization has its own security policies; these policies are followed by the employees according to their roles and responsibilities in the organization. Every person has a role or different roles that contain duties, and also a process cycle that he/she must follow to complete his/her tasks. So it is the same business concept used in Microsoft Dynamics AX.

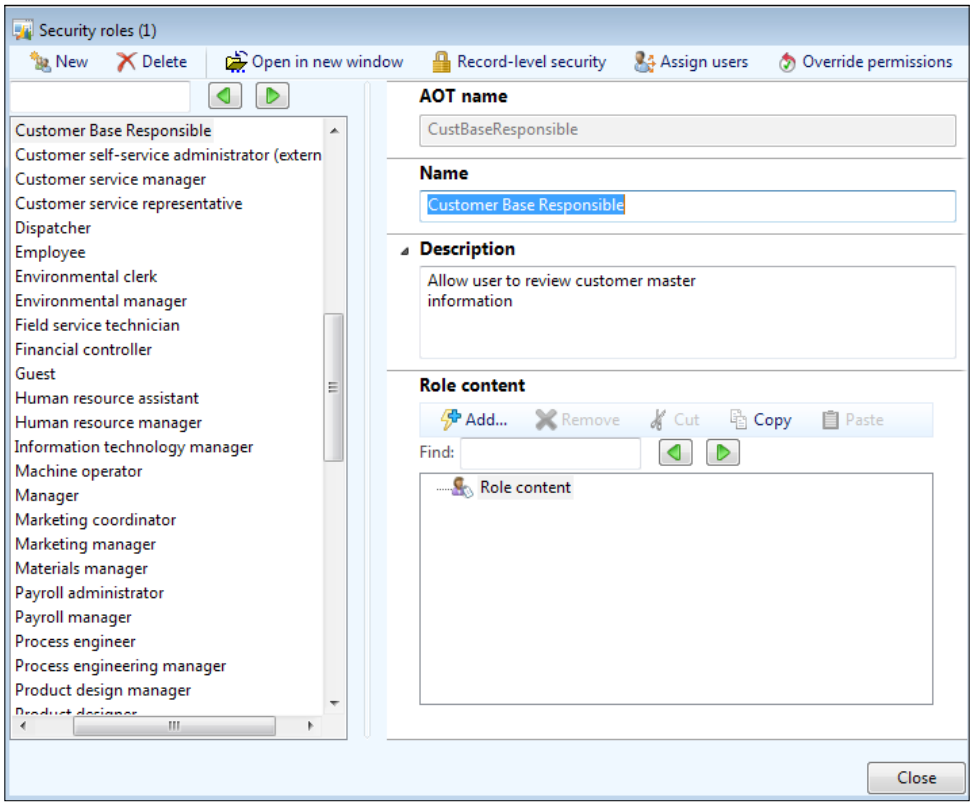
The following screenshot illustrates the hierarchy of the security concept within Microsoft Dynamics AX 2012:



- **Security policy**: This is a set of security roles that control the working environment
- **Security role**: This represents the scope of work for every person in the organization
- **Duties**: These are responsibilities to perform tasks for a specific business objective or process cycle, and they contain a set of application access privileges
- **Privileges and permissions**: These are a group of access rights granted to the users that require them

How to do it...

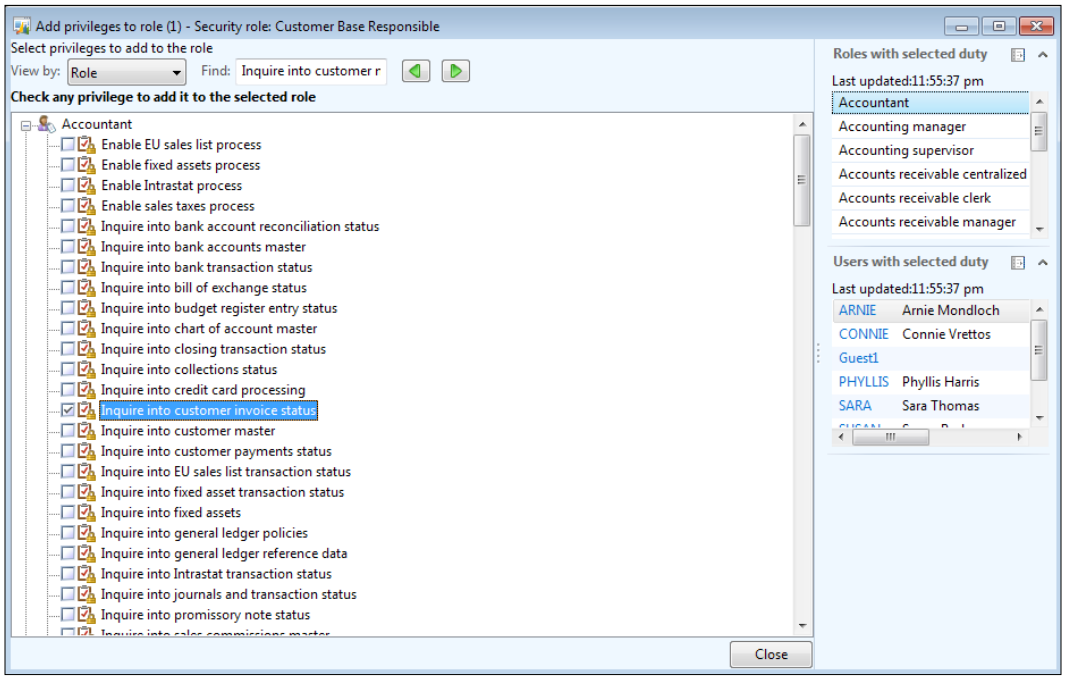
The following screenshot shows the security role form; on the left-hand side of the form, you will find a list of all the predefined security roles, and if you select one of the security roles, you will notice the related information at the center of the form. You can also drag-and-drop the security roles from the left-hand side of the form, into the **Name** field.



We are going to take a look at the steps to add a security role:

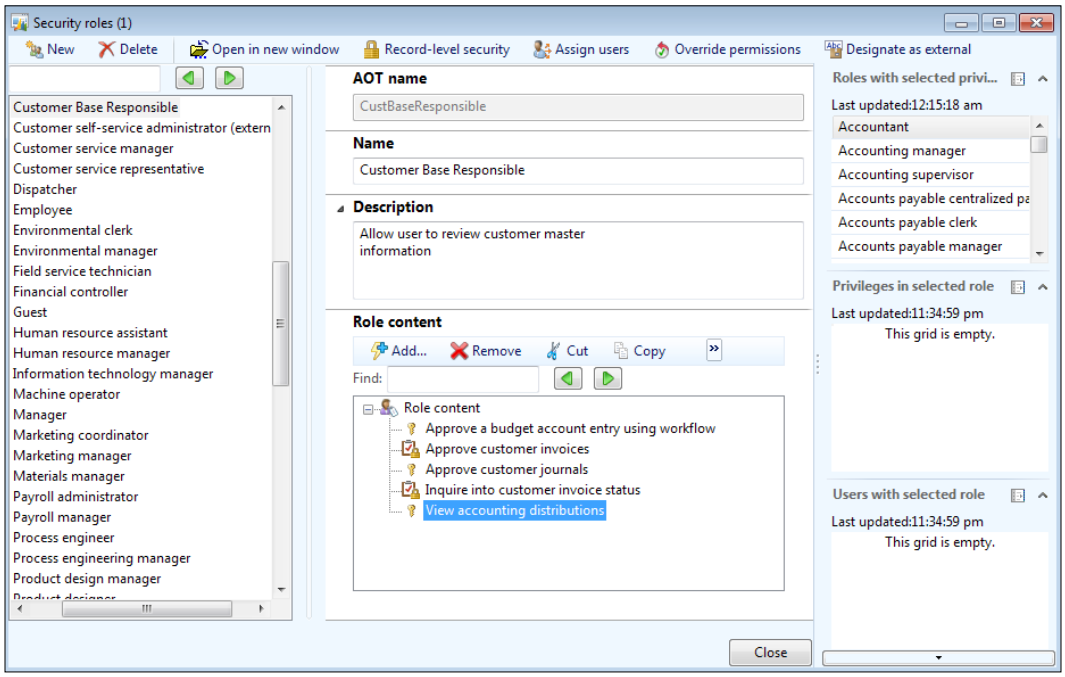
1. Go to **System administration** | **Setup** | **Security** | **Security roles**.
2. In the upper left-hand corner of the security roles form, click on **New**.
3. Enter the required role in the **AOT name** field.
4. Enter the name of the role in the **Name** field.

5. Enter a description for this role in the **Description** field.
6. Click on the **Add button** at the center of the form.



7. A new window appears; change **View by field** to **Role** to view the list of roles from the drop-down list.
8. You can type the role's name in the **Find** field to search for it.
9. Mark the selected role required by checking the checkbox on the left-hand side of the role name.

10. Click on **Close** to close the form and return to the main form.



11. You will notice that the role has been added to the **Role center section** on the main screen.
12. Click on **Close** to close this form.

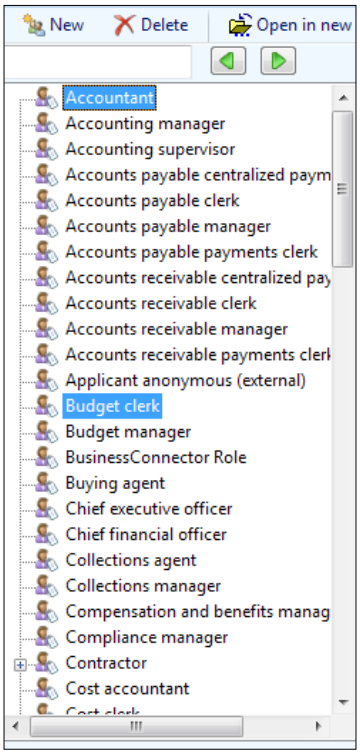
How it works...

We interacted with the **Application Object Tree** (**AOT**) by entering the role name in the **AOT name** field without interacting directly with the **AOT,** so the changes we made to the role form will also be made to the role node on the AOT.

There's more...

You can add a sub role to an existing role with the drag-and-drop function.

1. Open the security roles form.
2. On the left-hand side of the window, click on the existing role desired, which is **Budget clerk** in this case.



3. Again on the left-hand side of the window, click-and-drag the required role and drop to the existing role **Accountant in this case.**
4. You will notice that new roles have been added to the **Role center** section in the form.
5. Click on **Close** to close the form.

Adding privileges to an existing role (Become an expert)

In this section, you will learn how to add privileges or a process cycle to an existing role in a simple way.

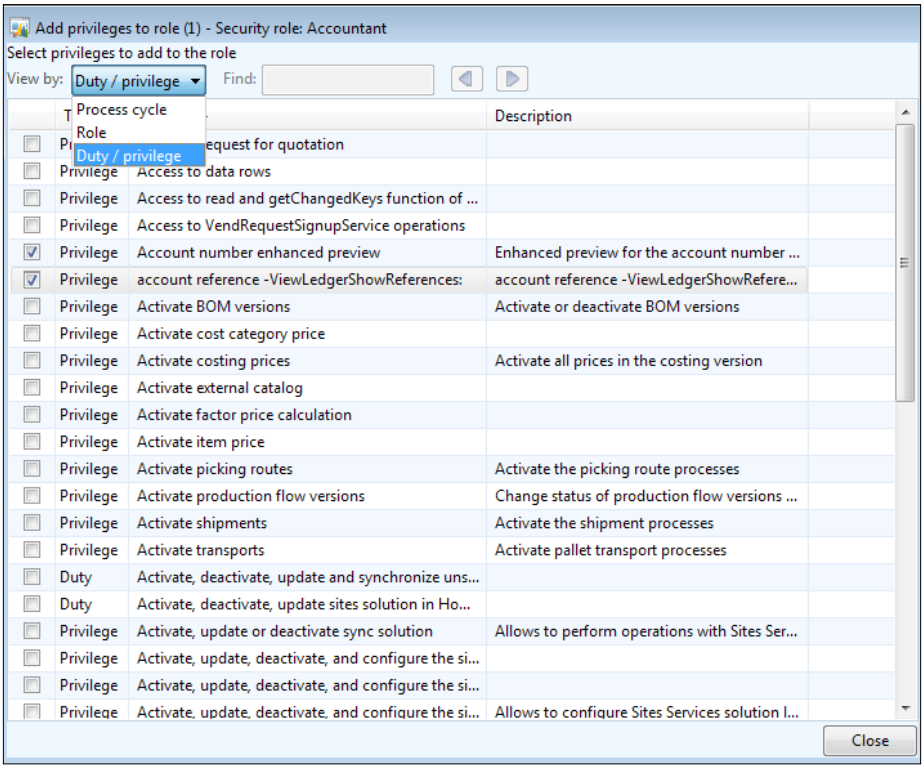
Getting ready

After adding the security role, we need to know how to set a specific duty, privileges, and process cycle related to the security role. We will follow the same steps, as in the previous section, with different changes that we will show now.

How to do it...

We are going to take a look at the steps needed to add privileges to an existing role:

1. Go to **System administration** | **Setup** | **Security** | **Security roles**.
2. Select the role from the list on the left-hand side of the form.
3. Click on the **Add button** at the center of the form under the **Role center** section.
4. After clicking on the **Add** button from the security role form, the window shown in the following screenshot will appear and show a list of privileges, process cycles, or roles according to your selection from the drop-down list.



5. A new window appears. Change the **View by** field to **Duty/privileges**, to view the list of privileges from the drop-down list menu.
6. You can also choose whether you want to add the duties or process cycle from the drop-down list menu.
7. After finishing your task, click on **Close** to close the form and return to the main form.

8. You will notice that the selected privileges from the previous step have been added in the **Role center section.**
9. Click on **Close** to close the form.

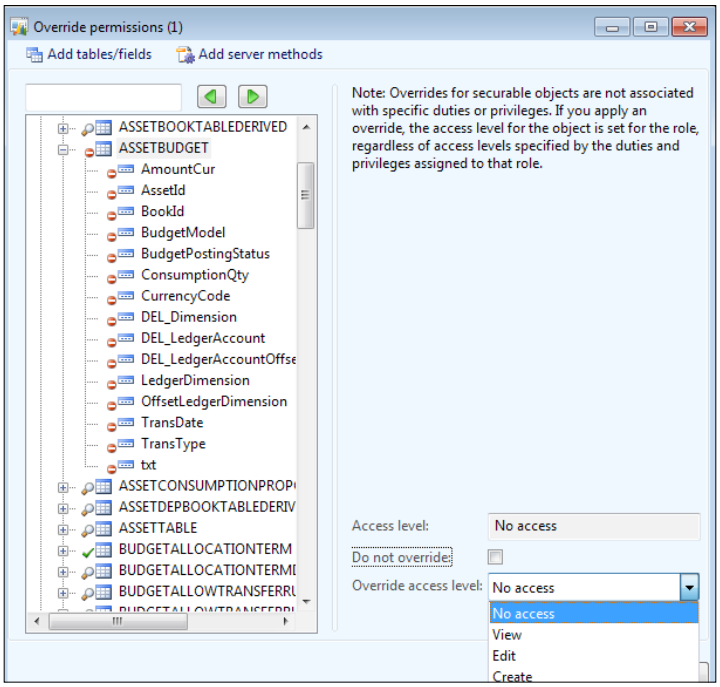
How it works...

We selected our existing role and added new privileges; we can add the duties and process cycle by following the same steps, and just change the view option from the drop-down list to get the full list of all the duties and process cycles.

There's more...

From the role form, you can override the permissions related to an existing role by clicking on **Override permissions** from the upper left-hand corner of the role form.

1. Click on the **Override permissions** button from the security form.
2. A new window appears. Select the required table or field from the left-hand side of the form.
3. Unmark the **Do not override checkbox**.
4. Select the access level from the drop-down list (**View**, **Edit**, **Create**, **Full control**).



5. Click on **Close** to close the form.

Securing the Organization Model (Become an expert)

We are going to look at how to limit a user's access according to the Organization Model hierarchy. We remember from earlier versions that if we needed to perform an action or access a table in a different company, we were required to log on to the different company; but with this new feature, **The Organization Model**, we can access any company or department in the company even if we are not logged on to it. So after creating the hierarchy of your organization, you will be able to perform multiple tasks for different companies at the same time.

We know from the previous sections that we can limit the users to access specific companies, but we can also limit the user according to the Organization Model hierarchy, and also according to specific departments or sections in all companies.

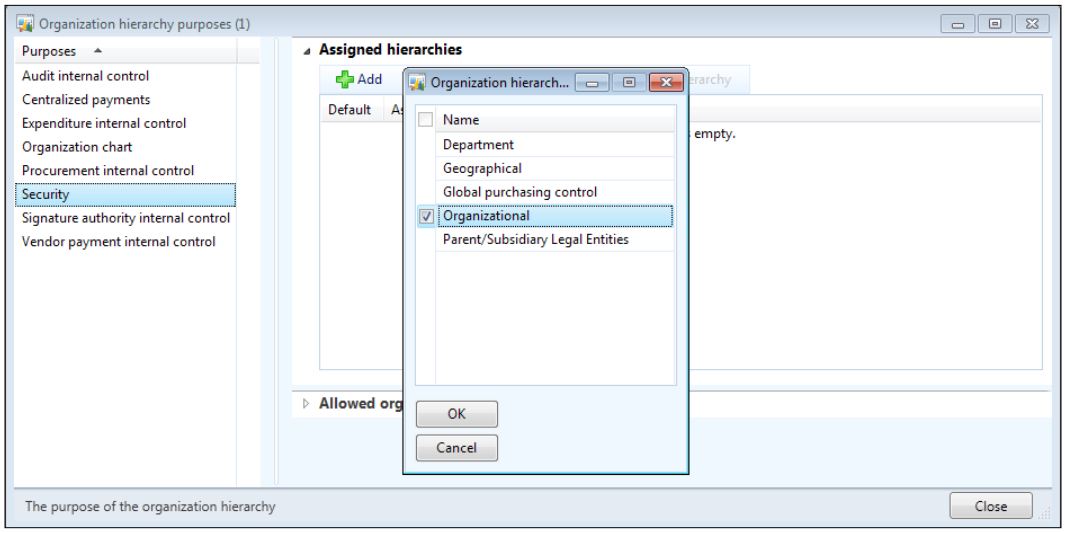
Getting ready

The Organization Model hierarchy must be set before doing the following steps.

How to do it...

We are going to take a look at the steps for granting user access to a specific department that was created previously, in the Organization Model hierarchy.

1. We first have to set the **Organizational** organization hierarchy for security purposes.
2. The following screenshot shows the organization hierarchy purpose, and how to set the security to the organizational model by clicking on the **Security** button on the left-hand side of the window. Click on the **Add** button that appears at the center of the window, and select **Organizational** from the list that appears.

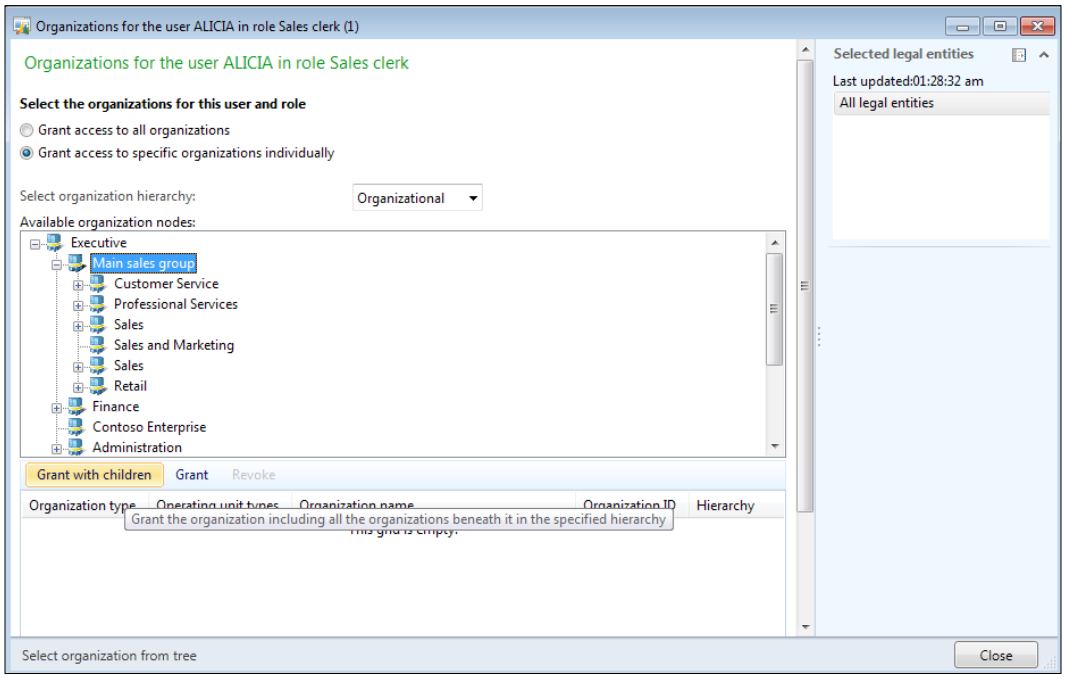


3. Go to **Organization administration** | **Setup** | **Organization** | **Organization hierarchies** | **Assign purpose** | **Organization hierarchy purposes**.
4. From the window that appears, click on **Security** from the left-hand side of the form.
5. Click on the **Add** button to add organizational hierarchy.
6. A new window appears; mark **Organizational.**
7. Click on **OK.**
8. Click on **Close t**o close the form.

You can limit the user's access to a specific department in the **Organizational** organization hierarchy, by performing the following steps:

1. Go to **System administration** | **Common** | **Users** | **Users**.
2. Select the user from the list.

3. Click on the **Assign organization button**.



4. Select **Grant access to specific organizations individually.**
5. Select the organization hierarchy to be **Organizational** from the drop-down list.
6. Select the appropriate department from the list provided.
7. Click on **Grant with children** to grant user access to the selected object and all subdepartments, or you can customize your selection by clicking on the **Grant** button for every department.
8. You will notice that the department to which access was granted has been added in the records in the lower part on the window.
9. Click on **Close** to close the window.

How it works...

We grant user access by using the Organization Model security, by first activating the security on the organizational model, and then granting the user access to specific organizations by changing the organization hierarchy to **Organizational**.

Developing an extensible data security policy (Become an expert)

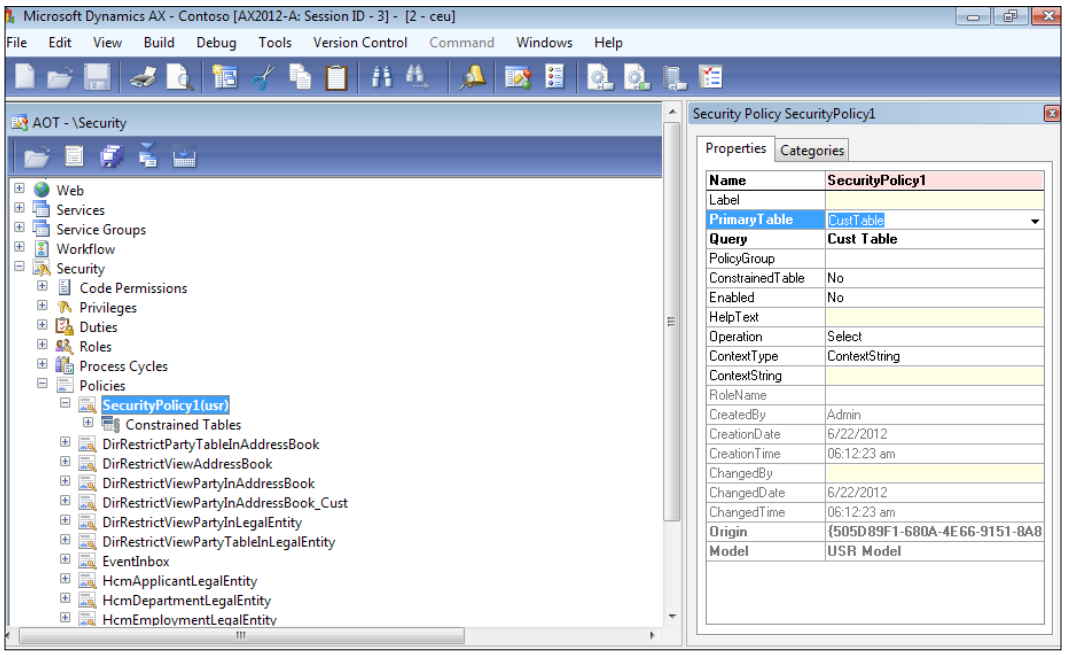
In this recipe, we will create data security policies based on primary tables, constrained tables, policy queries, and policy context.

Getting ready

The extensible data security policy is used to secure data in shared tables, by creating policies based on data that a contained in different tables. We can say that extensible data security is the development of the record-level security that was available in earlier versions, but in case you also need to use record-level security, Microsoft has kept this feature available in AX 2012 just for supporting backward compatibility for upgrades to the latest version.

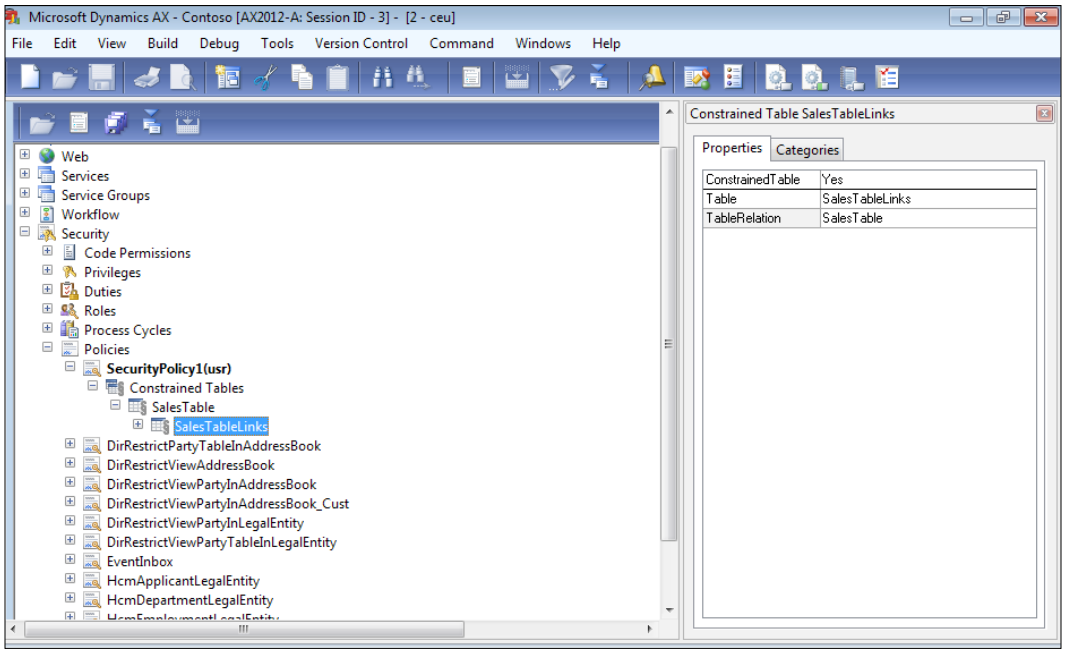
How to do it...

The following screenshot shows AOT after selecting the security node and policy node, to create a new security policy and assign a primary table to the created policy.



We are going to take a look at how to create the data security policy using the AOT:

1. Open the AOT window by pressing *Ctrl + D*.
2. Go to the **Security** node and expand it.
3. Right-click on the **Policies node** and select new security policy.
4. Enter a name for the new security policy that was created.
5. From the right-hand side of the window, select a table from the table property, **Cust Table**.
6. From the **Query property**, enter or select a query name.
7. Expand the security policy you created, and right-click on the **Constraints Tables subnode.**
8. The following screenshot shows how to assign constrained tables on the new security policy that was created.



9. From the menu that appears, select **New | Add table**.
10. From the right-hand side of the form, select the table from which you want to select the constraints **SalesTable**.
11. Select the table relation from the **TableRelation** field.

12. Expand the **Table** node, right-click the **RelatedTables** node, and select **New** | **Add table**.
13. Repeat the process for each constraint that you want to add to the policy.
14. Click on the **Save button** to save all your work.
15. Close the AOT window by clicking on **Close**.

How it works...

From the previous steps, we created a new security policy on AOT and made a primary relation with a specific table, Customer Table, and then related the created security policy to one or more constraint tables.

There's more...

Here you will find some definitions that will help you to get familiar with these concepts:

- **Constrained table**: A constrained table is the table or tables in a given security policy from which data is filtered or secured, based on the associated policy query. For example, in a policy that secures all sales orders based on the customer group, the Sales Order table would be the constrained table. Constrained tables are always explicitly related to the primary table in the policy.
- **Primary table**: A primary table is used to secure the content of the related constrained table. For example, in a policy that secures all sales orders based on the customer group, the Customer table would be the primary table.
- **Policy query**: A policy query is used to secure the constrained tables specified in a given, extensible data security policy. This query will return data from a primary table that is then used to secure the contents of the constrained table.
- **Context**: A policy context is a piece of information that controls the circumstances under which a given policy is considered to be applicable. If this context is not set, then the policy, even if enabled, is not enforced.

  Contexts can be of two types—role contexts and application contexts. A role context enables policy application based on the role or roles to which the user has been assigned. An application context enables policy application based on the information set by the application.

For more sources on this topic, refer to *Developing Extensible Data Security Policies*, White Paper published in *January, 2011* at `http://microsoft.com/dynamics/ax`.

Deploying changes in Enterprise Portal using AOT (Become an expert)

In this recipe we will apply the changes that occurred on AOT, and these changes will then be put into effect on the Enterprise Portal.

Getting ready

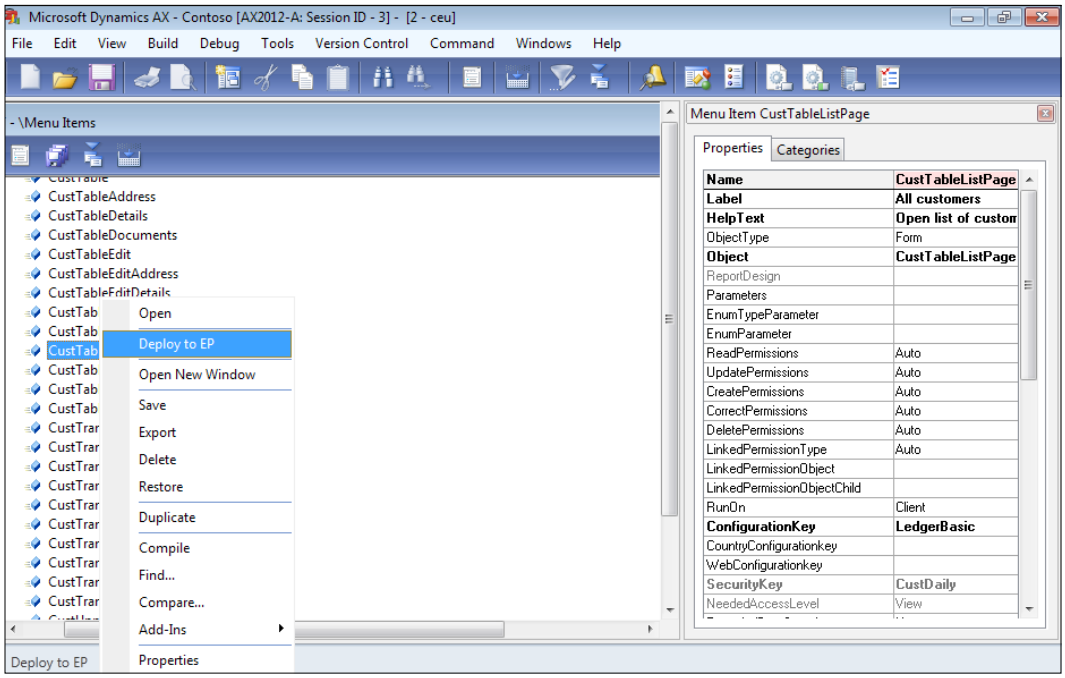
After finishing your work on AOT, you might want to see all these changes on the Enterprise Portal. Saving the work that you made on AOT will not affect the Enterprise Portal or SharePoint, so with this simple deployment, you can enable programmers to deploy all the changes with one simple form.

How to do it...

To deploy all the changes made on AOT to the Enterprise Portal, select the form you want to publish on the Enterprise Portal and perform the following steps:

The screenshot that follows the steps shows the AOT window. Expand the **Menu Items** section and click on the **Display** node to select the required table to publish on the Enterprise Portal.

1. Go to the **Menu Item** section associated with the form that you want to deploy from the AOT.
2. From **Menu Items**, select **Display**.
3. Go to the required table, **CustTable**.
4. Right-click on the required table.
5. From the menu displayed, click on **Deploy to EP.**



6. Click on **OK** to update the list.

How it works...

From the previous steps, we selected the table that is required to be published on Enterprise Portal, by opening the AOT form and expanding the menu items. We then selected the **Display** node to show all the tables and selected the table that is required to be deployed or published on Enterprise Portal.

Creating permissions in AOT (Become an expert)

In this section, we will take a look at how to create permissions that can be assigned to specific security roles using the AOT form.

Getting ready

We will learn how to create permissions for a new form and configure the necessary setup needed to finish this task in a simple way.

You first have to know the hierarchy of the permissions; each permission includes the following:

- Read
- Update
- Create
- Correct
- Delete

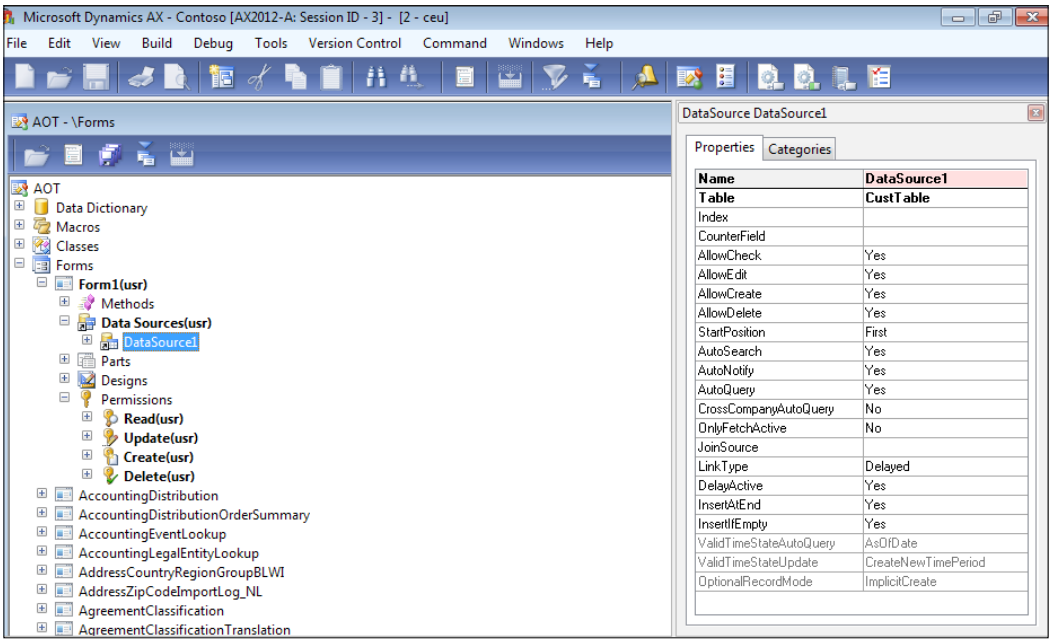
Read permission is the weakest permission and Delete is the strongest one, and Create permission includes Update and Read permissions.

Note that the descriptions of the permissions are initial descriptions, which are inferred by the system, so you can change and edit the initial values if necessary.

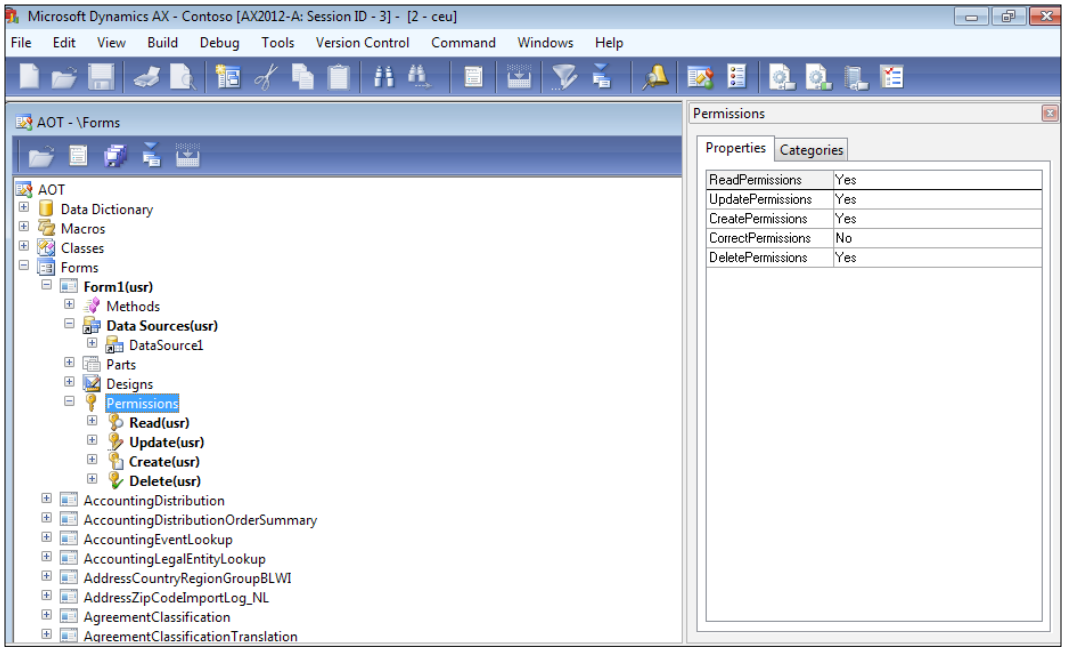
How to do it...

We are going to look at how we can create permissions for a new form and link it to the menu by performing the following steps:

1. The following screenshot shows the AOT form. Expand the forms node and create a new form. It also shows the set of permissions assigned to this form:

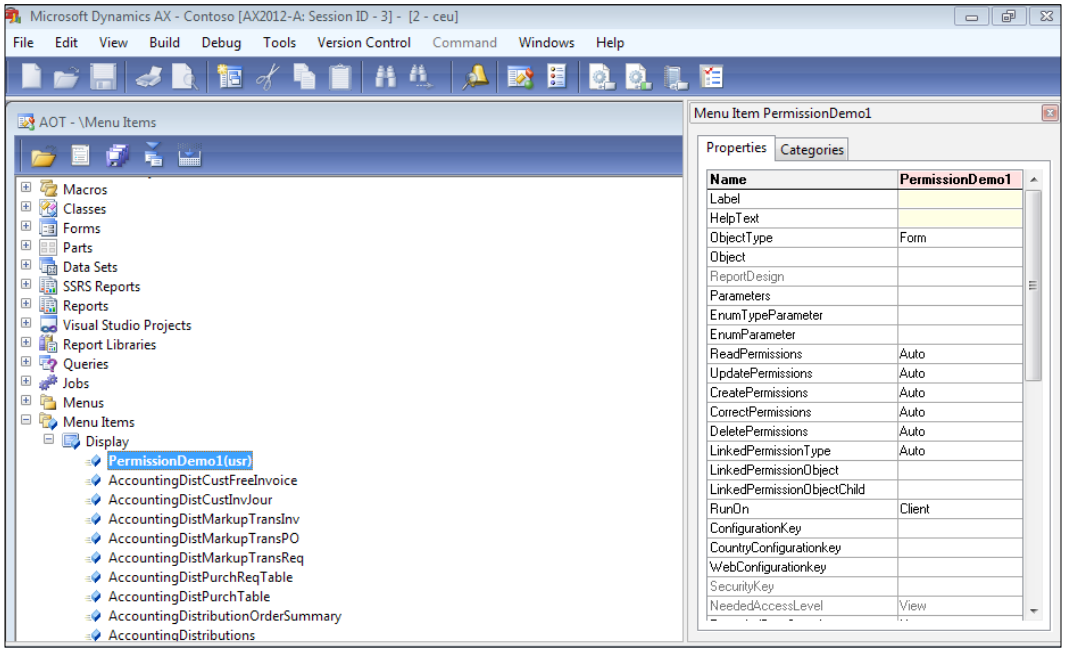


2. Open the AOT window by pressing *Ctrl + D*.
3. Expand the application object tree.
4. Right-click on the **Forms node** and create a new form.
5. Enter the name of the form as `Form1`.
6. Expand the new form.
7. Right-click on **Data Sources** and create a new data source.
8. From the right-hand side of the window, select a table to be the data source, namely **CustTable**.
9. Expand the **Permission node.**
10. The following screenshot shows the permission sets that are already generated by the system; you can also adjust the properties for every permission from the right-hand side of the form.



11. From the right-hand side of the window, make sure that the **CreatePermissions**, **UpdatePermissions**, **ReadPermissions**, and **DeletePermissions** property fields are set to the default value of **Yes**.
12. Only the **CorrectPermissions property** field is set to the default value of **No**.
13. Go to the **Menu Item node**.

The following screenshot and steps illustrate how to block permission sets from the form that was created, so the permission set will not be available to the selected users as a privilege on the menu item privilege node.



1. Right-click on **Display** and select **New** to create a new display menu item.
2. Enter a name for the new permission in the **Properties** tab, at the right-hand side of the window.
3. From the right-hand side of the window, make sure that the access-level properties are set to **Auto**.
4. Go to the **UpdatePermissions property** field, and change the value from **Auto** to **No**.
5. Click on **Save** to save your work.
6. Click on **Close** to close the AOT window.

How it works...

We defined the permissions, assigned to the new form created, that are required to access the menu item, and we changed the permissions to allow Read and Update access on the form, and also read-only access when a user wants to access the form from the menu item.

There's more...

- **Auto**: It means that the set of permissions will be available for the users to select the privileges on the menu item. For example, if the **UpdatePermissions** property is set to **Auto**, the permission set under **Form** | **Permissions** | **Update** will be available to select users for privileges under **AOT** | **Security**.
- **Access level**: It identifies a permission set under **AOT** | **Forms** | **Form1** | **Permissions**, or under similar nodes on other AOT elements.

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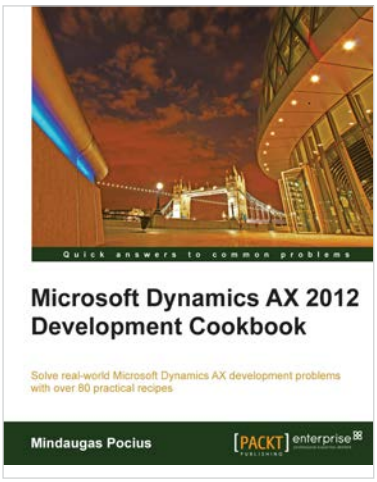

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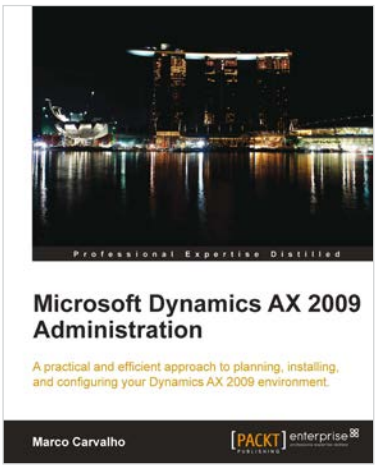

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